Spectral indices and incidental amplitudes in pixel-based Bayesian component separation: a tempered sampler, the reference frequency of a flat prior, and an honest error for the profile likelihood

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Abstract

In pixel-based Bayesian component separation a few spectral indices share the data with a very large number of per-pixel amplitudes. We study what three common ways of removing the amplitudes do to the index. (1) A Metropolis move that proposes an index, redraws the amplitudes and accepts on the χ^2 of the new pair is, on its own, a pseudo-marginal kernel for the profile posterior; inside a Gibbs sweep that also redraws the amplitudes it becomes Monte Carlo within Metropolis and, for small steps, converges to the profile posterior raised to the power $1/T$, with $T = \frac{1}{2} + 1/(4c_k) \simeq \sqrt{\pi k/8} + \frac{1}{2}$ for k amplitudes. The index error from such a chain is too wide by $0.79\,k^{1/4}$. We derive the law and its form for several moves per amplitude draw and for inexact stored amplitudes, and confirm it against an exact-kernel calculation and simulated chains within their Monte Carlo errors of 0.4 percent. (2) Moving the frequency at which a flat amplitude prior is stated multiplies the index prior by an exponential with rate $n \ln(\nu'_{\rm ref}/\nu_{\rm ref})$, n the number of amplitudes. For one component the index is shifted by $\ln(\nu_{\rm ref}/\nu_{\rm piv})$ times the index variance carried by one amplitude, a shift that does not average down over pixels, changes sign across a pivot frequency set by the bands and by the prior on the CMB, and, when the CMB amplitude is constrained, has opposite signs for references of 23 and 30 GHz with the WMAP and Planck LFI bands. (3) The profile likelihood has an exactly unbiased score, but its curvature is too sharp by $\sqrt{1 + \mathsf{T}/\mathsf{J}}$, where T/J is the inverse mean squared signal-to-noise ratio of the amplitudes; the natural plug-in sandwich correction is biased the wrong way, and a debiased form gives nominal coverage over four decades of T/J. For one component and a uniform noise pattern a zero-mean amplitude prior with a free scale is the same estimator; otherwise it is biased, by up to 0.6 standard deviations in controlled tests and by one standard deviation in simulations of the configuration below. We then apply the estimators side by side to the WMAP nine-year K to V polarization maps with their dense noise covariances, a ΛCDM prior on the CMB, four regions defined by geometry, and the brightest emission and Tau A masked. On the same maps, regions and noise model the polarized synchrotron index at $|b| > 30°$ is -3.15 with the profile likelihood, -3.17 with a flat amplitude prior stated at 23 GHz and -2.70 with the same prior stated at 30 GHz. The change of convention moves the index by 0.07 to 0.48 in the four regions, 4.8 to 16.4 statistical standard deviations, and the first-order form of the identity predicts every move, too high by 8 to 33 percent. With a free CMB amplitude the flat-prior marginal has no optimum at high latitude in either convention: the index runs to the edge of the allowed range, -1.6, where the profile likelihood gives -3.04. The zero-mean prior lands 3.6 standard deviations from the profile optimum at high latitude. A white prior on the CMB in place of the ΛCDM covariance, with nearly the same total variance, makes the error bar too small by 9 to 22 percent, by an exact first-order calculation that we give. With the mask no region is misfitted, errors from the scatter of the score over patches of sky agree with the model's, and the index does not depend on the angular scales or the band subset that inform it beyond the noise; it moves by up to 0.11 with the prior on the CMB and by 0.05 if the measured bandpasses are ignored. The χ^2 is 2.5 to 3.4 standard deviations low, which as a statement about the released covariances is noise overstated by 1.3 to 2.2 percent. An earlier version of this analysis, on regions reconstructed from a published index map and without a mask, had

regions that did not fit, an index that depended on angular scale, and one region whose index a single source, Tau A, moved by 0.08, six of its statistical errors.

1 Introduction

Bayesian component separation in the pixel domain fits, in every pixel, the amplitude of each sky component, and for the whole sky or for large regions a small number of spectral parameters that say how each component scales with frequency. In the Gibbs samplers that the Planck, BeyondPlanck and Cosmoglobe analyses are built on (Eriksen et al., 2008; Svalheim et al., 2023; Watts et al., 2023), the two kinds of parameter are updated in turn. The amplitudes are many and each is poorly determined; the spectral indices are few and are what the cleaning of large-scale polarization, and therefore the optical depth to reionization, depends on. The published polarized synchrotron indices from the same public maps span -2.9 to -3.3 depending on region, band set and method (Dunkley et al., 2009; Fuskeland et al., 2014; Svalheim et al., 2023; Weiland et al., 2022), which is larger than their quoted errors.

This paper is about the statistical step that sits between the maps and such an index: how the amplitudes are removed. That step has three places where it can go wrong without any error message, and we treat them in turn.

The sampler (Sect. 3). A move that proposes a new index, re-solves the amplitudes and accepts on the χ^2 of the new pair looks like a way to move both together. We show that on its own it is a valid pseudo-marginal kernel for the profile posterior, that the amplitude step of the Gibbs sweep turns it into Monte Carlo within Metropolis, and that the chain then converges to the profile posterior raised to the power $1/T$ with $T = \sqrt{\pi k/8} + \frac{1}{2}$ for k amplitudes. The index error bar from such a chain is too wide by $0.79\,k^{1/4}$, a factor 16 for the 1.9×10^5 amplitudes of an $N_{\rm side} = 64$ analysis, with an acceptance rate that stays near one half.

The prior (Sect. 4). With flat priors on the amplitudes, the standard choice, the index posterior contains the volume of amplitude space. This is known to bias indices (Eriksen et al., 2008; Dunkley et al., 2009; Alonso et al., 2017). We show that it is exactly equivalent to an exponential prior on the index whose rate is the number of amplitudes times the logarithm of the reference frequency at which the amplitudes are defined, give the resulting shift in closed form, and show that its sign flips between the two reference frequencies in common use, 23 and 30 GHz, for the WMAP and LFI bands when the CMB amplitude is constrained (with a free CMB amplitude both pull the index flat). The shift does not shrink as pixels are added.

The error bar (Sect. 5). The profile likelihood has neither problem and its score is exactly unbiased, but its curvature understates the error where the amplitudes are faint. We give the correction in closed form, show that the obvious way to evaluate it on real data is biased in the wrong direction, give the version that is not, and show its coverage over four decades of the controlling ratio. A zero-mean amplitude prior with a free scale turns out to be the same estimator when the noise pattern is uniform, and a biased one when it is not.

Sect. 6 applies the estimators side by side to the WMAP nine-year K to V band polarization maps with their released dense noise covariances, on regions defined by geometry with the brightest emission masked, so that the difference between their answers is the difference between the estimators. It then lists what else moves the index on these maps, by how much, and why we stop short of quoting one; an earlier choice of regions, which an independent referee showed to be wrong, is part of that account. Sect. 7 states where the recommended estimator stops being enough, and Sect. 8 describes the implementation and its tests. All results are for maps-only analyses at low resolution with amplitudes that enter linearly; we do not treat the time-ordered data.

2 Model, notation and the index targets in use

2.1 Data model

The data are maps d_b in bands $b = 1, \ldots, n_\mathrm{b}$, each a vector over elements p (pixels, and Stokes parameters in polarization), in thermodynamic units. The parametric model is

$$d_b = \sum_c m_{b,c}(\beta)\, a_c + n_b, \qquad n_b \sim \mathcal{N}(0, \mathsf{N}_b), \tag{1}$$

with one amplitude $a_c(p)$ per component c and element, a mixing coefficient $m_{b,c}$ that is the component spectrum averaged over the bandpass of band b, and spectral parameters β that are shared by many elements: one value for the sky, one per region, or a field at lower resolution than the amplitudes. Stacking bands and components, $d = \mathsf{M}(\beta)\, a + n$ with a of length k. We write

$$\mathsf{A} = \mathsf{M}^\mathsf{T}\mathsf{N}^{-1}\mathsf{M}, \qquad \hat{a} = \mathsf{A}^{-1}\mathsf{M}^\mathsf{T}\mathsf{N}^{-1}d, \qquad \mathsf{P} = \mathsf{N}^{-1} - \mathsf{N}^{-1}\mathsf{M}\mathsf{A}^{-1}\mathsf{M}^\mathsf{T}\mathsf{N}^{-1}, \tag{2}$$

all functions of β, so that

$$\chi^2(\beta, a) = \chi^2_\mathrm{min}(\beta) + (a - \hat{a})^\mathsf{T}\mathsf{A}\,(a - \hat{a}), \qquad \chi^2_\mathrm{min}(\beta) = d^\mathsf{T}\mathsf{P}d. \tag{3}$$

For the power-law component that carries the index, $m_b(\beta)$ is the band average of $(\nu/\nu_\mathrm{ref})^\beta$ in brightness temperature, converted to thermodynamic units, so that the amplitude is the brightness temperature at the reference frequency ν_ref. The reference frequency is a convention: it changes the meaning of a and not the model.

The amplitudes are what statisticians call incidental parameters (Neyman & Scott, 1948): their number grows with the data, and the parameters of interest, the indices, are few. Everything in this paper is a consequence of how the incidental parameters are removed.

2.2 Four ways to remove the amplitudes

Joint posterior with flat amplitude priors. With $p(a) \propto 1$ the joint posterior is $p(\beta)\exp[-\chi^2(\beta, a)/2]$ and integrating Eq. (3) over a gives the index marginal

$$\mathcal{L}_\mathrm{flat}(\beta) \propto \exp\!\left[-\tfrac{1}{2}\chi^2_\mathrm{min}(\beta)\right] \det \mathsf{A}(\beta)^{-1/2}. \tag{4}$$

A Gibbs sampler that draws $a \mid \beta$ with its fluctuation and then moves β on χ^2 at fixed a has exactly this marginal for β, whether or not it mixes well enough to show it (checked numerically and re-derived independently, E012). The determinant is the volume of amplitude space allowed at a given β. It is an implicit prior on the index and the subject of Sect. 4.

Profile likelihood. Maximizing over the amplitudes instead of integrating gives

$$\mathcal{L}_\mathrm{prof}(\beta) \propto \exp\!\left[-\tfrac{1}{2}\chi^2_\mathrm{min}(\beta)\right], \tag{5}$$

the spectral likelihood of Stompor et al. (2009) (their Eq. 9; their Eq. 10 is the marginal form). As a Bayesian target it is the marginal under the amplitude prior $p(a \mid \beta) \propto \det \mathsf{A}(\beta)^{1/2}$, a Jeffreys prior for the amplitude block conditional on the index. It is not the marginal under the joint Jeffreys prior of (a, β), which depends on a.

Gaussian amplitude prior. With $a \sim \mathcal{N}(\mu, \mathsf{S})$ the marginal keeps the form of Eq. (4) with $\mathsf{A} \to \mathsf{A} + \mathsf{S}^{-1}$, the prior mean entering χ^2_min, and a factor $\det \mathsf{S}^{-1/2}$ that matters when S is itself inferred. We use two cases. A zero-mean prior with one free scale per index region, $a_\mathrm{s}(p) \sim \mathcal{N}(0, \tau_r^2)$ for p in region r, with τ_r maximized jointly with the index (a prior flat in $\ln \tau_r$ needs a lower cutoff to be proper, since the likelihood tends to a constant as $\tau_r \to 0$). And, for intensity, a prior whose mean and scale are tied to a template, with hyperparameters inferred. Under a change of reference frequency τ_r rescales by $(\nu'_\mathrm{ref}/\nu_\mathrm{ref})^{\beta_r}$, and maximization is invariant under that rescaling, so the first case does not depend on the reference frequency.

Re-solving the amplitudes inside the index move. A fourth route is procedural: propose β', solve for new amplitudes at β', and accept on the χ^2 of the new pair. What this samples depends on how the new amplitudes are produced, and is the subject of Sect. 3.

2.3 Information and score

Let $\mathsf{M}_r = \partial\mathsf{M}/\partial\beta_r$ for index parameter r. For data that follow Eq. (1) with true amplitudes a, the profile score $s_r = -\frac{1}{2}\,\partial\chi^2_{\min}/\partial\beta_r$ has zero mean and covariance

$$\mathrm{Cov}(s_r, s_{r'}) = \mathsf{J}_{rr'} + \mathsf{T}_{rr'}, \qquad \mathsf{J}_{rr'} = (\mathsf{M}_r a)^{\mathsf{T}}\mathsf{P}\,(\mathsf{M}_{r'} a), \qquad \mathsf{T}_{rr'} = \mathrm{tr}\Big(\mathsf{A}^{-1}\mathsf{M}_r^{\mathsf{T}}\mathsf{P}\,\mathsf{M}_{r'}\Big)\,, \tag{6}$$

while the expected curvature of $\frac{1}{2}\chi^2_{\min}$ is J alone (derivation in App. A). These statements are for flat priors on every amplitude; with a Gaussian prior on a component they hold on average over realizations of that component drawn from the prior, with its precision included in A. J is the Fisher information on the indices with the amplitudes profiled out. T does not depend on the signal: it is the part of the score variance that the noise in the fitted amplitudes contributes. For one power-law component, diagonal noise with weights $w_{pb} = 1/\sigma^2_{pb}$ and one region, these reduce to

$$\mathsf{J} = \sum_p u_p V_p, \qquad \mathsf{T} = \sum_p V_p, \qquad u_p = a_p^2 \sum_b w_{pb} m_b^2, \qquad V_p = \mathrm{Var}_{wm^2}(\ln\nu)_p, \tag{7}$$

with u_p the squared signal-to-noise ratio of amplitude p and V_p the variance of $\ln\nu$ over the bands under the weights $w_{pb}m_b^2$: the lever arm.

3 What the re-solve move converges to

3.1 The move

The move we analyse is natural, and a version of it was the spectral-index step of the per-pixel path in the development version of Commander4 that we started from (option `resolve_chisq` in our branch). Given the current (β, a):

1. propose β' from a symmetric random walk;
2. draw new amplitudes from their conditional at the proposed index, $a' \sim \mathcal{N}\big(\hat{a}(\beta'), \mathsf{A}(\beta')^{-1}\big)$;
3. accept (β', a') with probability $\min\{1, \exp[-(\chi^2(\beta', a') - \chi^2(\beta, a))/2]\, p(\beta')/p(\beta)\}$, with p the index prior.

The Gibbs sweep then continues with the amplitude group, which redraws $a \sim \mathcal{N}(\hat{a}(\beta), \mathsf{A}(\beta)^{-1})$, and possibly repeats the index move n times before that. The intent is to move the index and the amplitudes together and so avoid the slow mixing of the conditional index step. Write $X = (a - \hat{a})^{\mathsf{T}}\mathsf{A}(a - \hat{a})$, so that $\chi^2 = \chi^2_{\min}(\beta) + X$ by Eq. (3). For an amplitude draw from the conditional, $X \sim \chi^2_k$ whatever β is.

3.2 Alone, the move is exact for the profile posterior

Because $\mathrm{E}[\exp(-X/2)] = 2^{-k/2}$ does not depend on β, the quantity $\exp[-\chi^2(\beta, a)/2]$ at a fresh conditional draw is an unbiased estimator of $\mathcal{L}_{\mathrm{prof}}(\beta)$ up to a constant. Steps 1 to 3, with the accepted amplitudes kept as part of the state, are therefore a pseudo-marginal Metropolis-Hastings kernel (Beaumont, 2003; Andrieu & Roberts, 2009). Its extended target is

$$\pi(\beta, a) \propto p(\beta)\, \exp\big[-\tfrac{1}{2}\chi^2_{\min}(\beta)\big]\, \det \mathsf{A}(\beta)^{1/2} \exp\Big[-(a - \hat{a})^{\mathsf{T}}\mathsf{A}\,(a - \hat{a})\Big]\,, \tag{8}$$

whose index marginal is the profile posterior and whose amplitude conditional is $\mathcal{N}(\hat{a}, (2\mathsf{A})^{-1})$: the amplitudes carry half their posterior covariance. So the move does have a target, the target is the profile posterior, and the determinant never enters. This is why chains run with this move have means close to the profile posterior. Under Eq. (8), $X \sim \mathrm{Gamma}(k/2, 1)$ with mean $k/2$.

3.3 The amplitude step of the Gibbs sweep breaks it

The amplitude group redraws a with the full conditional covariance A^{-1}, which resets X to a χ^2_k variate with mean k. That is the conditional of the flat-prior joint posterior and not of Eq. (8). The two kernels share no invariant distribution. With one index move per amplitude draw, both the current and the proposed estimate are fresh at every index move, and the composite is Monte Carlo within Metropolis in the terminology of Andrieu & Roberts (2009), following O'Neill et al. (2000) and Beaumont (2003). Andrieu & Roberts (2009) note that the target is typically not the invariant density of such a chain and that the existence of an invariant distribution is not obvious in general; later work bounds the distance between the noisy and the exact chain without giving the stationary law (Alquier et al., 2016; Medina-Aguayo et al., 2016). Here it can be written down in the limit of small proposal steps.

3.4 The stationary law: a tempered profile posterior

With one move per amplitude draw the index chain is Markov on its own, with acceptance probability

$$\alpha(D) = \mathrm{E}\big[\min\{1, e^{-(D+Z)/2}\}\big], \qquad D = \chi^2_{\min}(\beta') - \chi^2_{\min}(\beta) - 2\ln\frac{p(\beta')}{p(\beta)}, \qquad Z = X' - X, \quad (9)$$

where X and X' are independent χ^2_k variates, so Z is symmetric with variance $4k$. In the limit of small proposal steps the chain is a diffusion with drift proportional to $\alpha'(0)$ and diffusion coefficient proportional to $\alpha(0)$, and its stationary density is

$$\pi(\beta) \propto \big\{p(\beta)\exp\big[-\tfrac{1}{2}\chi^2_{\min}(\beta)\big]\big\}^{1/T}, \qquad T = \frac{1}{2} + \frac{1}{4c_k}, \qquad c_k = \mathrm{E}\big[e^{-Z/2};\, Z > 0\big] = 2^{-(k+1)}\binom{k}{k/2}. \quad (10)$$

The formula is exact for every k and every shape of $\chi^2_{\min}$ under four conditions: the amplitude draws are exact conditional draws; the k amplitudes redrawn by the Gibbs step are the ones that enter the acceptance; the proposal is symmetric with a fixed covariance; and the proposal steps are small against the stationary width. $T(0) = 1$ recovers Metropolis. For large k,

$$T = \sqrt{\frac{\pi k}{8}} + \frac{1}{2} + O(k^{-1/2}), \quad (11)$$

so the index posterior is the profile posterior with its width multiplied by $\sqrt{T} \simeq 0.79\, k^{1/4}$. The stationary acceptance rate is $\frac{1}{2} + c_k$, close to one half and independent of the step size, which is why such a chain looks healthy.

Equation (10) does not need χ^2_k noise. For any symmetric acceptance noise Z whose density f_Z is flat over a few units around zero, $T \simeq \frac{1}{2} + 1/(8 f_Z(0))$, and for Gaussian noise

$$T \simeq \frac{1}{2} + \sqrt{\frac{\pi}{8}}\,\sigma_E, \qquad \sigma_E = \tfrac{1}{2}\,\mathrm{sd}(Z), \quad (12)$$

with σ_E the standard deviation of the noise in the estimated energy difference, equal to $1/\sqrt{2}$ times the standard deviation of the χ^2 at a fresh amplitude draw. The leading term is the effective temperature that Ball et al. (2003) derived for the sign rule (accept if and only if the noisy energy difference is negative; their Eq. 7), and Bardenet et al. (2017) obtain, without assuming Gaussian noise, a temperature proportional to the noise with a slightly different constant (their Sect. 6.3). That a Metropolis chain on a noisy energy difference is over-dispersed is noted by Nicholls et al. (2012), and Ceperley & Dewing (1999) give the exact correction for Gaussian noise of known variance. The Metropolis rule on the same noisy difference adds the $\frac{1}{2}$. Equation (12) is the form to use when the stored amplitudes are not the drawn ones, as below.

Table 1: Variance of the stationary index distribution of the composite chain in units of the profile posterior variance, one index move per amplitude draw, for $\chi^2_{\min} = \beta^2$ and exact χ^2_k refreshes. Columns: Eq. (10); its leading term; the stationary vector of the exact kernel on a grid of 1201 points, with the acceptance function computed by convolution of the χ^2_k density; and a simulation of 10^4 chains with a proposal step of $0.1\sqrt{T}$ (E035, independent review). The grid sits 0.2 to 0.3 percent below the law because of the finite step, and the simulation scatters about it within its Monte Carlo error. Controls: with no refresh, or with a refresh from Gamma$(k/2, 1)$, the variance is 1.000 ± 0.001 at $k = 4$, 8 and 16.

k	T, Eq. (10)	$\sqrt{\pi k/8}$	exact kernel	simulation	acceptance
10	2.532	1.982	2.524	2.525 ± 0.008	0.621
100	6.782	6.267	6.769	6.751 ± 0.024	0.538
1 000	20.32	19.82	20.29	20.26 ± 0.07	0.511
10 000	63.17	62.67	63.07	63.30 ± 0.23	0.503
100 000	198.67	198.17	198.35	199.39 ± 0.72	0.500
200 000	280.75	280.25	280.31	280.22 ± 1.01	0.500

Several index moves per amplitude draw. After the first accepted move the retained X is no longer fresh: it is biased low, and the next proposal has to beat it. In the large-k limit the accepted moves are the lower records of the sequence of fresh X values, the j-th proposal after a redraw is accepted with probability $1/(j+1)$, and the stationary law remains a power of the profile posterior with temperature $T_n = R_n T_1$,

$$R_n = \frac{H_{n+1} - 1}{\sqrt{\pi}\, N_\varphi(n)}, \qquad N_\varphi(n) = \sum_{j=1}^{n} j \int \varphi(t)^2 \left[1 - \Phi(t)\right]^{j-1} \mathrm{d}t, \tag{13}$$

where H_n is the harmonic number, φ and Φ are the standard normal density and distribution function, and $N_\varphi(n)$ is the sum over proposals of the expected density at the running minimum. $R_2 = 5/6$ exactly, and numerically $R_n = 0.833$, 0.647, 0.543, 0.467 and 0.357 for $n = 2$, 5, 10, 20 and 100: more index moves narrow the law only slowly, asymptotically as $(\ln n)^{-1/4}$ in width, and the $k^{1/4}$ scaling is untouched.

The repair that is exact and useless. Redrawing the amplitudes with half their covariance, $a \sim \mathcal{N}(\hat{a}, (2\mathsf{A})^{-1})$, is the conditional of Eq. (8) and makes the composite exact for the profile posterior. Its acceptance rate at fixed index is $2I_{1/3}(k/2, k/2)$, with I the regularized incomplete beta function, which decays as $e^{-0.059k}$: 6×10^{-4} at $k = 100$. The useful repairs evaluate Eq. (5) or Eq. (4) directly in the acceptance, which costs one extra log-determinant per proposal at most (Sect. 8).

3.5 Tests

Table 1 and Fig. 1 compare Eq. (10) with the exact kernel and with simulated chains. The law was derived twice, the second time by an independent reviewer who was given the algorithm and not our derivation. The two derivations agree, the reviewer's exact-kernel calculation reproduces every row of our own toy within 1.4 standard errors, and the controls return the profile posterior exactly, so the excess is caused by the full-covariance redraw and by nothing else. For proposal steps that are not small the kernel is no longer reversible with respect to a power of the profile posterior, but for $k \geq 10^4$ the variance stays within 8 percent of T at any step size (0.918 T for steps of four stationary widths or more; 0.907 T at $k = 100$).

The same toy reproduces, within 5 percent, four earlier measurements of the width of chains run with this move on simulated maps, with k from 450 to 3072 and one or four moves per amplitude draw (measured 3.9, 5.7, 3.2 and 4.4 profile widths; toy 4.0 to 4.1, 5.7, 3.0 and 4.3; E002, E005).

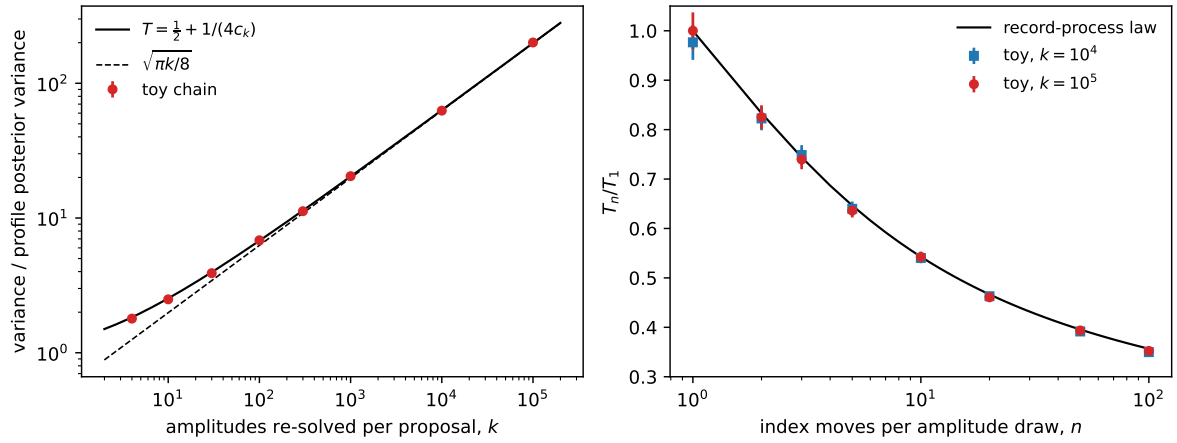


Figure 1: The stationary law of the re-solve move inside a Gibbs sweep. Left: temperature of the tempered profile posterior against the number of amplitudes re-solved per proposal, for one index move per amplitude draw: Eq. (10) (line), its leading term (dashed) and the toy chain (points). Right: the ratio T_n/T_1 for n index moves per amplitude draw: the record-process law of Eq. (13) (line) and the toy at $k = 10^4$ and 10^5 (points).

Real maps. On nside-64 intensity maps (17 bands from Haslam to Planck 857 GHz, 38 165 unmasked pixels, five components, so $k = 190\,825$) a chain with 20 index moves per amplitude draw, proposing the synchrotron and dust indices jointly with steps of 5.6 and 93 profile widths, was run twice. The widths were 17.0 and 16.4 profile widths for the synchrotron index (about 8 percent error each) and 16.4 and 16.6 for the dust index (2 percent each) (`E014`). For small steps Eq. (13) gives $\sqrt{R_{20}T_1} = 11.3$; a toy chain with two indices and the recorded step sizes gives 13.1 and 13.6. The measured widths are 20 to 30 percent above that. We can account for the excess after the fact, and say so plainly: the account uses one number measured from the log of the same chain. This code path stores the per-pixel amplitudes as spherical-harmonic coefficients to $\ell_{\max} = 3N_{\rm side} - 1$ and evaluates the χ^2 of the acceptance on the round-tripped maps. The round trip is not exact on the HEALPix grid, so the stored amplitudes differ from the drawn ones. The log records the log acceptance ratio of every proposal, and at the first move of each iteration both amplitude sets are fresh: the scatter of those 3200 values is 682 ± 9 where exact draws give $\sqrt{k} = 437$, a factor 2.44 ± 0.06 in variance, and a direct calculation of the round-trip residual on the same inputs gives 2.69 ± 0.18. With the factor 2.44 in Eq. (12) the toy gives widths of 15.9 and 16.6 and an acceptance rate of 0.075 against the measured 0.074 and 0.077. The same storage residual has a gradient in the indices that predicts shifts of the chain means of $+3.1$ profile widths for the dust index and -0.1 for the synchrotron index, where $+3.0 \pm 0.5$ and -2.7 ± 2.0 were measured. The forward test, a run of the same configuration with the amplitudes stored as maps, predicted to give 13.1 and 13.6 widths and no shift, has not been made.

3.6 What this means for existing chains

A chain run with this move reports index uncertainties that are too wide by $\sqrt{T}$, a factor of order ten at map-level k, with a mean close to the profile posterior. It does not sample the flat-prior joint posterior that the rest of the Gibbs sweep assumes. The statement applies to the move as defined in steps 1 to 3 with exact conditional draws. We met it in the development version of Commander4. In Commander3 we read the code and did not run it: the index samplers used for the BeyondPlanck analysis hold the amplitudes fixed during the Metropolis step, which is a valid conditional update and not this move, while a later module (`sample_specind_mh`) proposes an index, re-solves the amplitudes by conjugate gradients and accepts on χ^2, which has the structure analysed here (`E009`). We do not know which published results, if any, were made with that module.

4 The flat amplitude prior, written in terms of the reference frequency

That the flat-prior marginal of Eq. (4) pulls spectral indices is known. Eriksen et al. (2008) found a global offset biased by more than 4σ under uniform priors and removed it with a Jeffreys prior on the index. Dunkley et al. (2009), fitting one polarized synchrotron index per $N_{\rm side} = 2$ pixel to the five-year WMAP maps with uniform priors on the CMB and synchrotron amplitudes defined at 23 GHz, found index posteriors centred on -2.6 in simulations, noted that the effect grows with the number of pixels that share an index and that a Jeffreys prior on the index does not remove it, and divided their prior by a factor estimated from a noise-only simulation. Alonso et al. (2017) cancel the determinant with a "volume prior", which in our reading makes their target the profile likelihood, and show biases of more than 3σ without it. Jeffreys or reference priors are used against prior-volume effects in several other analyses (Dickinson et al., 2009; Planck Collaboration X, 2016; Jew & Grumitt, 2020), and Andersen et al. (2023) describe the marginal and the profile ("ridge") index samplers of Commander, noting that the two differ by the determinant term. What has not been written down, as far as we can find, is how large the pull is in closed form and what sets its sign. Both follow from one identity.

4.1 An exact identity

Change the reference frequency of the power-law component from $\nu_{\rm ref}$ to $\nu'_{\rm ref}$. The mixing column of every amplitude in index region r is multiplied by $(\nu_{\rm ref}/\nu'_{\rm ref})^{\beta_r}$, so $\mathsf{M}' = \mathsf{M}\,\mathsf{D}$ with D diagonal, $\chi^2_{\rm min}$ is unchanged, and $\det \mathsf{A}' = \det \mathsf{A} \prod_r (\nu_{\rm ref}/\nu'_{\rm ref})^{2 n_r \beta_r}$, with n_r the number of power-law amplitudes in region r. Therefore

$$\ln \mathcal{L}_{\rm flat}(\beta; \nu'_{\rm ref}) = \ln \mathcal{L}_{\rm flat}(\beta; \nu_{\rm ref}) + \sum_r n_r\, \beta_r \ln \frac{\nu'_{\rm ref}}{\nu_{\rm ref}}, \tag{14}$$

for any number of components, any bandpasses and any noise covariance. Moving the frequency at which a flat amplitude prior is stated is the same as multiplying the index prior by an exponential, $e^{\lambda\beta}$ with $\lambda = n \ln(\nu'_{\rm ref}/\nu_{\rm ref})$: 34 e-folds per unit of index for 128 amplitudes moved from 23 to 30 GHz. The profile likelihood and the zero-mean prior with a free scale do not change.

4.2 The shift and its pivot

Let $g_r = -\frac{1}{2}\,\partial \ln\det \mathsf{A}/\partial\beta_r$ be the gradient of the volume term. To first order the maximum of $\mathcal{L}_{\rm flat}$ sits at the profile maximum plus

$$\delta\beta = \mathsf{H}^{-1} g, \qquad \mathsf{H} = \mathsf{J} + \nabla\nabla\, \tfrac{1}{2} \ln\det \mathsf{A}, \tag{15}$$

where H is the expected curvature of the flat-prior marginal itself: the profile information plus the curvature of the volume term. For one power-law component with delta bands and diagonal noise, $-\frac{1}{2}\ln A_p = \beta \ln \nu_{\rm ref} - \frac{1}{2}\ln \sum_b w_{pb}\, \nu_b^{2\beta}$, so

$$g = \sum_p \ln \frac{\nu_{\rm ref}}{\bar\nu_p}, \qquad \ln \bar\nu_p = \frac{\sum_b w_{pb}\, \nu_b^{2\beta} \ln \nu_b}{\sum_b w_{pb}\, \nu_b^{2\beta}}, \qquad \nabla\nabla\, \tfrac{1}{2} \ln\det \mathsf{A} = 2 \sum_p V_p = 2\,\mathsf{T}, \tag{16}$$

where $\bar\nu_p$ is the frequency at which amplitude p is best measured: the geometric mean of the band frequencies under the weights (inverse noise variance times squared spectrum) that also define the lever arm V_p in Eq. (7). When the amplitudes of a region share a noise pattern, $\bar\nu_p = \nu_{\rm piv}$ and, with $\sigma_\beta^2 = 1/\mathsf{J}$ the profile variance of the region index,

$$\boxed{\delta\beta = \ln\!\left(\frac{\nu_{\rm ref}}{\nu_{\rm piv}}\right) \frac{n\,\sigma_\beta^2}{1 + 2\,\mathsf{T}/\mathsf{J}} \;\longrightarrow\; \ln\!\left(\frac{\nu_{\rm ref}}{\nu_{\rm piv}}\right) \sigma_1^2 \quad (\mathsf{T} \ll \mathsf{J})} \tag{17}$$

where $\sigma_1^2 = n\sigma_\beta^2$ is the index variance that a single amplitude carries, and T/J is the inverse mean squared signal-to-noise ratio of the amplitudes (Sect. 5). The boxed form is for one component. With further free amplitudes the curvature of the volume term is no longer $2\mathsf{T}$ and can be negative, so that H is smaller than J and the flat-prior marginal may have no interior optimum at all; one of our fits to real maps, for a band set without K band and with a free CMB amplitude (on the earlier, unmasked regions described in Sect. 6), ends at the edge of the allowed range (E040). Three things follow.

The shift does not average down. Adding amplitudes at fixed depth leaves σ_1 and T/J unchanged, so $\delta\beta$ stays the same while the posterior width falls as $n^{-1/2}$. In units of the posterior width the pull grows as $\sqrt{n}$: this is the observation of Dunkley et al. (2009) that the effect increases with the number of pixels per index, and it is the Neyman-Scott structure (Neyman & Scott, 1948). More data of the same kind make it worse. So does a finer pixelization of the same data: for $\mathsf{T} \ll \mathsf{J}$, four times as many amplitudes for the same region information is four times the shift.

The sign is set by the convention. The index is pulled steep if the reference lies below the pivot and flat if above. A flat prior is neutral only when stated at the pivot.

The pivot depends on the other components and on their priors. A component with a free amplitude and a spectrum that does not depend on the index (the CMB, dust with fixed parameters) contributes nothing to g directly, but projecting it out replaces the weights w_{pb} by the projected weight matrix $\tilde{\mathsf{W}}_p = \mathsf{W}_p - \mathsf{W}_p\mathsf{F}(\mathsf{F}^\mathsf{T}\mathsf{W}_p\mathsf{F} + \mathsf{S}^{-1})^{-1}\mathsf{F}^\mathsf{T}\mathsf{W}_p$, with F the spectra of the other components and S their prior covariance. The pivot becomes a generalized mean under $\tilde{\mathsf{W}}_p$, which has negative off-diagonal elements and can lie outside the range of the bands. With a free CMB amplitude under a flat prior the index information sits in the lowest band and the pivot moves to or below it. With a CMB prior the size of the signal it moves back.

For several regions, several components and dense noise, g and H are a vector and a matrix, and Eq. (14) gives the difference between two references,

$$\hat{\beta}_{\mathrm{flat}}(\nu'_{\mathrm{ref}}) - \hat{\beta}_{\mathrm{flat}}(\nu_{\mathrm{ref}}) \simeq \mathsf{H}^{-1}\left[\, n_r \ln(\nu'_{\mathrm{ref}}/\nu_{\mathrm{ref}}) \,\right], \tag{18}$$

in which the regions are coupled: a large region with many amplitudes drags its neighbours through the off-diagonal elements of H^{-1}. We test this form on simulations and on real data in Sect. 6. Equation (15) was corrected by an independent review of our first derivation, which had J in place of H; the two agree where $\mathsf{T} \ll \mathsf{J}$, and a direct test with the reference placed close to the pivot gives a measured shift of 1.01 and 1.03 times Eq. (15) at $\mathsf{T}/\mathsf{J} = 0.03$ and 0.11 where the uncorrected form is off by 5 and 16 percent (E039, independent review). For shifts that are not small against the range over which the curvature changes, neither is accurate (Fig. 2, right).

4.3 Sizes at the depth of the public polarization data

Table 2 gives the sizes on a realistic polarized sky at the white-noise depth of the WMAP and LFI maps. For the synchrotron-only model the pivot of the five lowest bands is 26.4 GHz, and references of 23 and 30 GHz, the two in use, sit on opposite sides of it: the same data give an index steeper than the truth by 0.06 in one convention and flatter by 0.06 in the other, one posterior standard deviation each way, for regions measured to 0.06. For regions measured to 0.16 the pull is 0.4 and 0.3. The difference between the two references follows Eq. (14): $\ln(30/23) \times 128 \times 0.059^2 = 0.118$ against a measured 0.123. Doubling and quadrupling the noise leaves the shift at a given σ_β unchanged, as Eq. (17) requires (-0.063 and -0.073 at 23 GHz for the class 0.03 to 0.1; E036). The profile likelihood on the same simulations is unbiased to 0.002 in the first class and 0.02 in the second (Sect. 5).

With free CMB and dust amplitudes under flat priors the pivot moves to 22 to 23 GHz. A flat prior stated at 23 GHz is then nearly neutral and one stated at 30 GHz pulls the index flat by 0.17 to 0.86. With a Gaussian CMB prior of 0.2 μK per element, the size of the CMB polarization signal at this resolution, the pivot and the shifts return to the synchrotron-only values, and so

Table 2: The flat-prior marginal against Eq. (17) on simulations at the white-noise depth of the WMAP and LFI polarization maps (E036). One index per $N_{\rm side} = 2$ pixel (128 Q/U amplitudes at $N_{\rm side} = 16$), exact one-dimensional posteriors on a grid, 100 noise realizations; truth: the Cosmoglobe DR1 polarized synchrotron map smoothed to 5° with $\beta = -3.1$. Regions are grouped by their profile index error σ_β. "Shift" is the mean of posterior mean minus truth. The predicted value is the high signal-to-noise limit of Eq. (17) with the class median of σ_β (in brackets) and the pivot of the band set; it is first order in the shift and uses one number for a class, so it degrades where the shift exceeds the posterior width.

model	σ_β class (median)	reference	shift	predicted	shift / posterior sd
synchrotron only,	below 0.03 (median 0.019)	23 GHz	−0.005	−0.006	−0.26
bands K, 30, Ka, Q, 44		30 GHz	+0.007	+0.006	+0.35
($\nu_{\rm piv} = 26.4$ GHz)		60 GHz	+0.037	+0.038	+1.9
	0.03 to 0.1 (0.059)	23 GHz	−0.061	−0.062	−0.94
		30 GHz	+0.062	+0.057	+1.01
		60 GHz	+0.341	+0.37	+6.1
	0.1 to 0.25 (0.161)	23 GHz	−0.40	−0.46	−2.4
		30 GHz	+0.34	+0.42	+2.5
free CMB and dust, flat priors,	0.03 to 0.1	23 GHz	+0.006		
adding V, 70, 353 GHz		30 GHz	+0.167		
	0.1 to 0.25	23 GHz	+0.058		
		30 GHz	+0.859		
the same with a CMB prior	0.03 to 0.1	23 GHz	−0.060		
of 0.2 μK per element		30 GHz	+0.063		

does the index error, which the flat CMB prior had inflated by a factor 1.63. The prior on the CMB amplitude matters more for the synchrotron index than the prior on the synchrotron amplitude does, as long as the latter is not flat.

An existing real-data comparison. An earlier experiment of ours (E022) evaluated both the flat-prior marginal and the profile likelihood on real WMAP Ka to V, LFI and 353 GHz polarization maps, in four regions reconstructed from the released index map of Svalheim et al. (2023) and without a mask (Sect. 6 explains why we no longer use those regions; the argument here is differential and does not depend on what they are), at a 30 GHz reference, for two noise models. In its high signal-to-noise limit Eq. (17) says the difference of the two estimates divided by $n\sigma_\beta^2$ is $\ln(30/\nu_{\rm piv})$ whatever the noise model. In the region reconstructed as the Spur the two noise models give shifts of +0.271 and +0.172 for $n\sigma_\beta^2 = 1.913$ and 1.216: a pivot of 26.04 GHz in both cases, to 0.2 percent, for a band set whose lowest band is at 28.4 GHz and with a free CMB amplitude. With a CMB prior of 0.19 μK the implied pivot is 29.5 to 30.2 GHz. For that band set the neutral reference frequency is therefore 26 GHz or 30 GHz depending on how the CMB is constrained. For WMAP K to V on the real maps it is 23.3 GHz with the ΛCDM prior on the CMB and 16.9 to 40.2 GHz, below the lowest band, with a flat one (Sect. 6). Sect. 6 tests the identity directly on the WMAP maps with their dense covariances.

5 Two targets without the volume term, and an honest error for the simpler one

5.1 The profile likelihood: unbiased score, curvature too sharp

For amplitudes that enter linearly with known noise, the profile score is exactly unbiased whatever the true amplitudes are (App. A). The amplitudes are fixed unknowns, no distribution is assumed for them, and nothing depends on the reference frequency. This is the functional errors-in-variables model with known error covariance. There the consistency of the slope is classical,

and so is the caveat that the variance of the estimator is not the one derived from the Fisher information matrix (Engel, 1985, and references therein; Fuller 1987 is the textbook); Stompor et al. (2009) remark of the profile curvature that it gives "useful, if not strictly precise" uncertainties. What follows is the form of that caveat for several components and dense noise, with its size. In the notation of Eq. (6) the score variance is $\mathsf{J}+\mathsf{T}$ while the curvature of $\frac{1}{2}\chi^2_{\min}$ is J, so the estimator has covariance

$$\mathrm{Cov}(\hat{\beta}) = \mathsf{J}^{-1}(\mathsf{J}+\mathsf{T})\,\mathsf{J}^{-1}, \tag{19}$$

a sandwich (Huber, 1967; White, 1982) whose middle term is known in closed form here. An interval built from the curvature of the profile likelihood, or from an MCMC chain run on Eq. (5), is too narrow by

$$\frac{\mathrm{sd}(\hat{\beta})}{\sigma_{\mathrm{curv}}} = \sqrt{1+\frac{\mathsf{T}}{\mathsf{J}}} = \sqrt{1+\frac{\sum_p V_p}{\sum_p u_p V_p}}, \tag{20}$$

the second form for one region and one component. The ratio T/J is the inverse of the lever-arm-weighted mean squared signal-to-noise ratio of the amplitudes. For a region measured through a few bright pixels it is negligible. For an index shared by many faint pixels it is not, however well the index itself is measured: this is why the profile likelihood was found unbiased but "up to two times too narrow" for a full-sky index at low signal to noise in our earlier simulations (E017), and why it is safe for the bright regions of the polarized sky (E036).

5.2 The error from one data set

On real data J has to be evaluated at fitted amplitudes, and here is a trap. The fitted amplitudes are the true ones plus noise with covariance A^{-1}, and

$$\mathrm{E}\big[\mathsf{J}(\hat{a})\big] = \mathsf{J}+\mathsf{T}\,: \tag{21}$$

the plug-in information estimates the score variance, not the curvature. A sandwich built as $\mathsf{J}(\hat{a})^{-1}[\mathsf{J}(\hat{a})+\mathsf{T}]\mathsf{J}(\hat{a})^{-1}$ is therefore narrower than the curvature width it was meant to widen. The consistent estimate is

$$\mathsf{J}_{\mathrm{c}} = \mathsf{J}(\hat{a}) - \mathsf{T}, \qquad \widehat{\mathrm{Cov}}(\hat{\beta}) = \mathsf{J}_{\mathrm{c}}^{-1}\,\mathsf{J}(\hat{a})\,\mathsf{J}_{\mathrm{c}}^{-1}, \tag{22}$$

and a region for which J_{c} is not positive has fitted amplitudes consistent with noise and no measured index. Both $\mathsf{J}(\hat{a})$ and T come from quantities already formed in the fit. Subtracting the expected noise contribution from a quantity built on noisy regressors is the classical correction of the measurement-error literature: the corrected score and corrected observed information of Nakamura (1990), and in astronomy the debiased sample variance of Akritas & Bershady (1996). What is specific here is the closed form of T for several components with dense noise, and that the uncorrected plug-in errs on the narrow side.

Figure 2 tests this over a ladder designed to reach the regime where it matters: many faint amplitudes sharing one index, with predictions written down before the run (E041). With the curvature width the standard deviation of the standardized error equals Eq. (20) within 0.054 in all 30 cells where the index is measured, up to 1.59 at $\mathsf{T}/\mathsf{J}=1.56$, and 68 percent intervals cover the truth as rarely as 49 percent of the time. With the plug-in sandwich they cover it 38 percent of the time. With Eq. (22) the standardized error has standard deviation 0.95 to 1.05 and the coverage is 0.65 to 0.72 in 29 cells and 0.635 in one; the mean standardized error is within 0.10 everywhere. Sect. 6 repeats the test of the error taken from each realization for several coupled regions, three components and the dense WMAP covariances, in a regime where T/J is small. Two limits of the recipe, found by an independent check: the first-order sandwich is about 3 percent narrow on average and up to 7 percent at $\mathsf{T}/\mathsf{J}\simeq 1$, and above $\mathsf{T}/\mathsf{J}\simeq 1.5$ the plug-in estimate is too noisy to use even when J_{c} is positive, so we quote an error only where $\mathsf{T}/\mathsf{J}_{\mathrm{c}}<1$. With a Gaussian prior on a component that does not depend on the index, such as the CMB, the score is unbiased over skies drawn from that prior and not for a sky whose variance differs from it; the first-order bias is given in App. A.

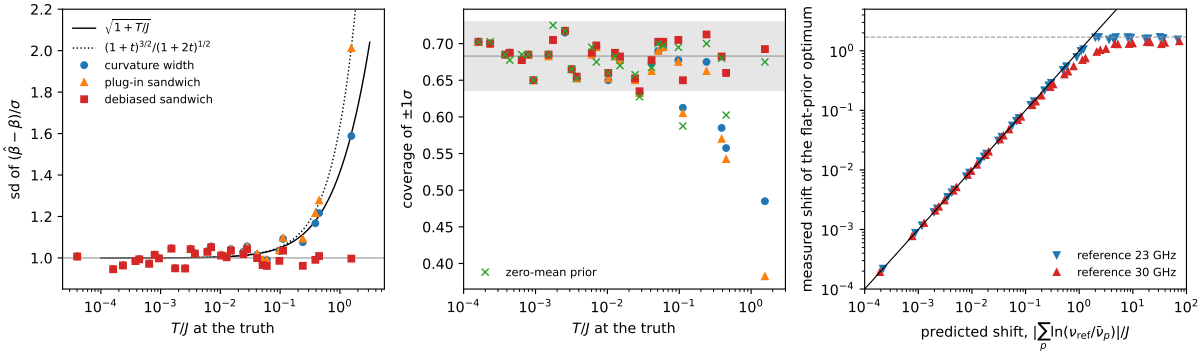


Figure 2: One index shared by $n = 32$ to 8192 heavy-tailed amplitudes at five bands, per-amplitude signal-to-noise ratio from 9 to 0.07, 400 noise realizations per cell; cells in which the index is measured to better than 0.25 (E041). Left: standard deviation of the standardized error of the profile optimum with three error estimates, each taken from the same data set as the estimate, against T/J at the truth. The curvature width follows Eq. (20) (line), the plug-in sandwich is narrower still (dotted: its expected factor), and the debiased sandwich of Eq. (22) stays at one. Centre: coverage of the $\pm1\sigma$ interval for the same three and for the zero-mean prior; the band is ±2 binomial standard errors around 0.683. Right: shift of the flat-prior optimum from the profile optimum against g/J, the high signal-to-noise limit of Eq. (15), at references of 23 and 30 GHz, all 40 cells; the dashed line is the edge of the allowed range.

5.3 The zero-mean prior with a free scale

The Bayesian relative of the profile likelihood is a zero-mean Gaussian amplitude prior with a variance per index region that is inferred with the index. For a noise pattern that is common to the amplitudes of a region, $w_{pb} = w_b$, the marginal likelihood depends on the data through $q(\beta) = \sum_p b_p^2/A$ only, with $b_p = \sum_b w_b m_b d_{pb}$ and $A = \sum_b w_b m_b^2$, and maximizing over τ gives

$$\ln \mathcal{L}_\tau(\beta) = \tfrac{1}{2}\, q(\beta) - \tfrac{n}{2} \ln q(\beta) + \text{const} \qquad (q > n), \tag{23}$$

a monotonic function of the profile log-likelihood $\frac{1}{2}q$. The optimum is the profile optimum, and the curvature is the profile's times $1 - n/q$, which is the plug-in form of $1/(1 + \mathsf{T}/\mathsf{J})$. In this case (one component, a common noise pattern, the scale maximized) the zero-mean prior with a free scale *is* the profile likelihood with its sandwich width: the prior learns from the scatter of the fitted amplitudes how much of them is noise. Simulation confirms the identity to the tolerance of the optimizer (E041).

When the noise pattern varies between the amplitudes, as it does on real maps, the two estimators part. The Gaussian prior is then a model for the amplitudes, real polarized amplitudes are far from Gaussian, and the bright ones carry whatever noise pattern they happen to have. In the ladder the resulting bias of the zero-mean prior is between -0.6 and $+0.2$ posterior standard deviations in the cells where the index is measured, with a sign that belongs to the amplitude configuration and not to the noise level, while its coverage stays within 0.59 to 0.73. On the realistic sky of Table 2 it was within 0.11 standard deviations except in one class at four times the real noise (-0.30; E036). A scale map in place of a single scale reduces it there. With three components, dense noise and a faint high-latitude region, the configuration of Sect. 6.3, it reaches one standard deviation in simulations and more on the maps (E044). We therefore take the profile likelihood with the debiased sandwich as the estimator of record and report the zero-mean prior beside it. What the prior offers in exchange is a generative model for the amplitudes, which a hierarchical model of an index field needs.

5.4 What each target is good for

Table 3 collects the results. Two remarks on scope. First, for an index *field* with a hierarchical prior the profile likelihood is not enough at low signal to noise: the profile maxima of faint regions

Table 3: The index targets of Sect. 2 for an index shared by many amplitudes, as established by the experiments cited.

target	centre	width	depends on ν_{ref}
flat-prior marginal (and Gibbs with flat amplitude priors)	shifted by $\ln(\nu_{\mathrm{ref}}/\nu_{\mathrm{piv}})\,\sigma_1^2$ (E036, E041)	curvature	yes: tilt $e^{n\beta \ln \nu_{\mathrm{ref}}}$
profile likelihood	unbiased (E036, E041)	curvature too narrow by $\sqrt{1+\mathsf{T}/\mathsf{J}}$; debiased sandwich calibrated (E041)	no
zero-mean prior, free scale per region	unbiased for a common noise pattern; biased otherwise, by up to 0.6 sd in the ladder and one sd in Sect. 6.3 (E041, E044)	calibrated where unbiased	no
re-solve move inside a Gibbs sweep	close to the profile optimum	too wide by $\sqrt{T}$, Eq. (10) (E035)	no

scatter, and a hierarchy reads that scatter as field variance (E012). A proper amplitude prior is then needed, and its specification becomes part of the model (Sect. 7). Second, the statements above are for amplitudes that enter linearly with known Gaussian noise. Beams, masks and a correlated-noise model that is itself uncertain change the numbers, not the structure.

6 Demonstration on the public polarization maps

This section does one thing: it applies the estimators of Sects. 4 and 5 to the same real maps, with the same regions, mask, bands and noise model, so that the difference between their answers is the difference between the estimators and nothing else. It does not deliver a polarized synchrotron index. Sect. 6.4 shows why not: at the statistical precision these maps offer, analysis choices that are independent of the estimator move the index by as much as its error or more.

6.1 Data and model

WMAP. The nine-year low-resolution ($N_{\mathrm{side}} = 16$) Stokes Q and U maps of the K, Ka, Q and V bands (Bennett et al., 2013) with their released dense inverse noise covariance matrices, which carry the time-correlated noise of the differential scan and de-weight the poorly constrained transmission-imbalance modes (Jarosik et al., 2007, 2011). We checked them as released (E038): symmetric to 10^{-9}, positive definite with no null space, a pixel noise of 2.0 to 2.2 μK in brightness temperature, and a few modes at multipoles 2 to 6 whose variance is 65 to 164 times the median. Watts et al. (2023) argue from a joint reprocessing that these matrices do not fully describe the large-scale uncertainty of the nine-year maps; we use them as released and test the sensitivity to the modes they flag.

Planck. The PR4 map at 353 GHz (Planck Collaboration Int. LVII, 2020) as the dust tracer. For it the per-pixel Q/U covariance of the map file is combined over the sub-pixels of each $N_{\mathrm{side}} = 16$ pixel, and the map is the corresponding weighted average.

Spectra. Each band enters through its measured bandpass. For WMAP every detector is given unit response to a thermal spectrum and the detectors and radiometers of a band are averaged with equal weights, which reproduces the per-radiometer effective frequencies of the WMAP team's routine to 0.005 GHz, and the team's correction for the drift of the band centres over

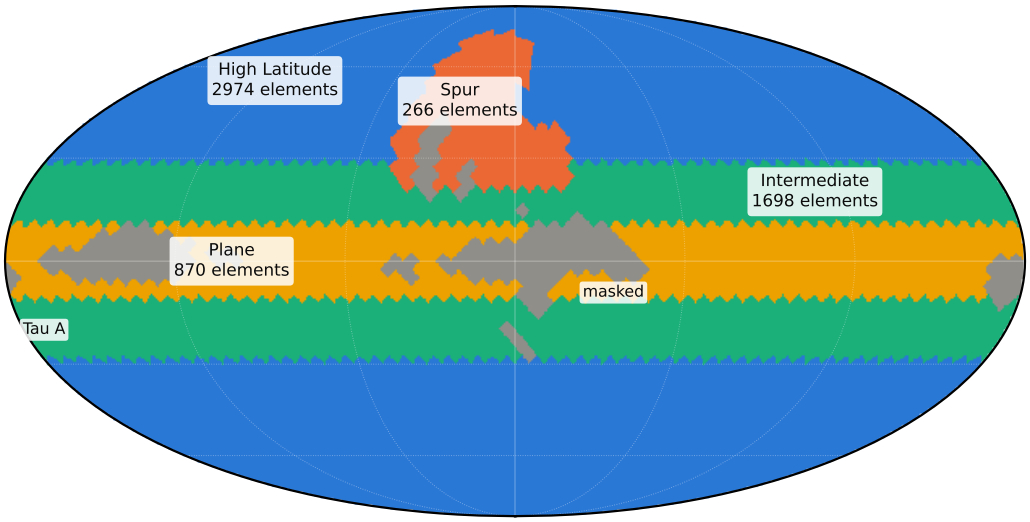


Figure 3: The four regions and the processing mask (E044), in Galactic coordinates with the Galactic centre in the middle. An element is one Stokes parameter of one $N_{\rm side} = 16$ pixel.

the mission (0.13 percent at K band) is applied. For a power law of index -3.1 the effective frequencies are 22.41, 32.66, 40.30 and 59.99 GHz for K to V and 28.09 GHz for the 30 GHz band. Delta bands at the nominal WMAP frequencies would steepen a K to Ka index by about 0.06 (a calculation, not a fit). The 353 GHz band cannot be a delta band: a modified blackbody with index 1.53 and temperature 19.6 K (Planck Collaboration XI, 2020) seen through the band is 1.12 to 1.14 times its value at 353.0 GHz, depending on which bolometers are counted, so a delta band overstates the dust at every lower frequency by that factor. We use the band of the polarization-sensitive bolometers.

Beams. The low-resolution maps are pixel averages of maps with beams from 0.88° (K) to 0.08° (353 GHz). A noise-free simulation with index -3.1 everywhere, each band seen through its own Gaussian beam, gives region indices biased flat by 0.005 to 0.011 for the WMAP bands (E038). We therefore bring every band to a common 1° window with an invertible operator on the $N_{\rm side} = 16$ grid, $\mathsf{S}_b = 1 + \mathsf{Y}\,\mathrm{diag}(f_\ell - 1)\,\mathsf{Y}^+$ with f_ℓ the ratio of the beam windows to multipole 47, transforming the data and the covariance together. In the simulation the residual bias is below 0.0025, and on the real maps the correction moves the four indices by 0.0068, 0.0025, 0.0085 and 0.0035 where the simulation predicted 0.0070, 0.0030, 0.0088 and 0.0041 (measured with the white CMB prior, on an earlier set of four regions and without a mask; E040).

Regions and mask. Four regions defined by geometry (Fig. 3): *High Latitude*, $|b| > 30°$; *Intermediate*, $10° \le |b| \le 30°$; *Plane*, $|b| < 10°$; and *Spur*, cut out of the first two, the northern part ($b > 20°$, $-45° < l < 65°$) of the pixels to which the released index map of Svalheim et al. (2023) gives its Spur value, which covers the North Polar Spur. A processing mask removes the 5 percent of pixels with the largest polarized amplitude in the Cosmoglobe DR1 synchrotron model at 30 GHz smoothed to 5°, which are the inner plane, the Fan region and the base of the Spur, and a disc of 7.5° around Tau A: 5808 of 6144 elements are kept, 2974, 266, 1698 and 870 in the four regions. Svalheim et al. (2023) describe a mask of this kind for their index sampling (the strongest 5 percent of the signal, and strong sources such as Tau A); ours follows that description and is not their file. Masked elements are removed exactly: the kept elements enter with the inverse of their own block of each covariance, the Schur complement of the full inverse covariance, formed after the window matching. An earlier version of this analysis used four regions reconstructed from the released index map without a mask; an independent referee showed that its "High Latitude" region held the inner Galactic plane and that Tau A dominated

the misfit of its "Plane" (Sect. 6.4), and those regions are not used for any statement below except where named.

Model. Synchrotron as a power law with one index per region and one amplitude per element (pixel and Stokes parameter); dust as the modified blackbody above with one amplitude per element under a flat prior; and the CMB with one amplitude per element. The prior on the CMB amplitudes is the ΛCDM covariance of E and B modes on the grid: spectra from CAMB (Lewis et al., 2000) for the Planck 2018 parameters (Planck Collaboration VI, 2020) with an optical depth of 0.054, through the working window and the pixel window, for multipoles 2 to 47, the span of the harmonic basis of the grid, plus a white floor of 0.01 μK. The pixel average of a ΛCDM sky has a further 0.11 μK per element from higher multipoles, nearly white, which this prior does not describe (independent review, E040); with the floor raised to that value the four indices change by 0.0006 or less and their errors by 0.3 percent. Two other choices are shown beside it, because the choice matters both for the index and for the pivot of Sect. 4: a white Gaussian prior of 0.2 μK per element, which we used first, and a flat prior, for which the profile score is unbiased whatever the CMB is. The white prior has nearly the right total variance (0.17 μK per element for ΛCDM) and the wrong shape: it understates the ΛCDM E-mode variance by a factor of 200 at the quadrupole and overstates it above multipole 10. With the mask, the prior on the kept elements is the kept block of that covariance.

Expectations. Expectations for every fit of this section were written down, with numbers, before the fit was run, and the simulations of the next subsection were read before any real map (E039, E040, E044). The fits on the present regions were not the first look at these maps: they follow a round on the earlier regions whose results we had seen, and the choice of the ΛCDM prior over the white one was made by a rule written three minutes before the first ΛCDM fit, after a referee had shown the white prior wrong in shape. The records mark what was written when, and record the expectations that failed; several did, and the text says so where it matters.

6.2 Validation on simulations with the real covariances

Before any fit to real data the chain was run on simulations with known indices: synchrotron amplitudes from the Cosmoglobe DR1 polarized synchrotron map, dust amplitudes from the 353 GHz map, a CMB drawn from the white prior, one true index per region chosen away from the centre of the allowed range (-3.15, -3.25, -2.85, -2.95), and noise drawn from the real covariances (E039). Six criteria were fixed in advance. The simulations use the four regions of the earlier version of this analysis as their geometry, and its names appear in Table 4 as labels only; nothing tested here depends on what the regions are.

Table 4 gives the outcome for the four regions at $N_{\mathrm{side}} = 8$, where the simulated sky is constant inside each pixel by construction. The profile optimum is unbiased to 0.07 standard deviations, its sandwich error is right to 6 percent, the 68 percent intervals cover the truth 69 percent of the time, and the reduced χ^2 at the optimum is 1.0006 with a scatter of 0.022. A spot check at the working resolution, $N_{\mathrm{side}} = 16$, with 12 realizations gives a pooled standardized error of mean $+0.03$ and standard deviation 0.99 and a reduced χ^2 of 0.998 ± 0.009. The zero-mean prior lands within 0.06 standard deviations of the profile optimum on these regions; on the fainter regions of the map fit it does not (Sect. 6.3). For this band set with the white CMB prior the pivot sits at 23 GHz: a flat prior stated there is neutral, and the same prior stated at 30 GHz moves the High Latitude index by 4.5 standard deviations. The single-region formula predicts that move for the large region and fails for the three small ones, which move by 1.7 to 5 times their own prediction: they are dragged by the large region through the shared noise. The coupled form, Eq. (18) with J^{-1}, is within 5 percent in all four. Two of our six criteria failed as written, one because it was worded wrongly (a lower bound of 1.0 on a ratio that scatters around one) and one because it was written for the single-region formula. In these regions T/J is below 0.02, so the errors taken

Table 4: Validation of the region-index estimators on simulations with the released WMAP noise covariances (E039): 200 noise realizations at $N_{\mathrm{side}} = 8$, the first 40 with every estimator. z is the error of the profile optimum in units of the sandwich error at the truth. The last block compares the difference between the flat-prior optima at 30 and 23 GHz with Eq. (17) applied to each region alone and with the coupled form, Eq. (18), evaluated with J^{-1} in place of H^{-1} (the two agree here, where T/J is below 0.02).

	High Latitude	Spur	Galactic Center	Plane
elements	1124	212	50	150
sandwich error at the truth	0.0153	0.0166	0.0108	0.0146
mean of z	-0.02 ± 0.07	0.00 ± 0.07	$+0.07 \pm 0.07$	$+0.01 \pm 0.07$
standard deviation of z	1.04	0.98	0.94	1.02
coverage of $\pm 1\sigma$	0.670	0.715	0.685	0.695
zero-mean prior minus profile, in σ	+0.03	0.00	+0.06	+0.04
flat prior at 23 GHz minus profile	−0.0008	0.0000	−0.0002	−0.0004
flat prior at 30 GHz minus profile	+0.0680	+0.0270	+0.0074	+0.0138
in units of σ	+4.5	+1.6	+0.7	+0.9
30 GHz minus 23 GHz, measured	+0.0687	+0.0270	+0.0076	+0.0142
predicted, each region alone	+0.0692	+0.0156	+0.0015	+0.0085
predicted, coupled, with J^{-1}	+0.0723	+0.0278	+0.0078	+0.0145

from each realization (Eq. 22) differ from the errors at the truth by less than a percent: with the errors taken from each of 40 realizations the standardized error has a standard deviation of 0.97 and the 68 percent coverage is 0.70, for the curvature, plug-in and debiased errors alike. The correction itself is tested in Fig. 2, not here. These simulations were first run on a numerical path that was later found to overwrite memory (Sect. 8); they were run again on a sound path and every number in Table 4 and in this paragraph was reproduced to the last printed digit (E043).

6.3 The same maps, several answers

Table 5 is the result of this section. The numbers in each column come from the same maps, the same regions and mask, the same bands, bandpasses and noise covariances. They differ only in how the amplitudes are removed.

No region is misfitted: the χ^2 per element and band is 0.577, 0.591, 0.572 and 0.614 in the four regions against an expectation of 0.598 (sampling errors 0.009, 0.030, 0.012 and 0.017). In total it is low. The minimum χ^2, with the penalty of the CMB prior inside it, is 16954 for 17420 degrees of freedom, 2.5 standard deviations below its expectation, and the deficit is in the two faint regions; 40 simulations of this fit at $N_{\mathrm{side}} = 8$ give a ratio of 1.002 ± 0.003. A fluctuation of that size has a probability of about one percent, and with a flat prior on the CMB, where the count of degrees of freedom is exact, the deficit is larger (below). The other reading is that the released covariances overstate the noise away from the plane, by 1.3 to 2.2 percent in rms if it is spread evenly, which would make every error below too large by that factor. We found no statement of the accuracy of the normalization of these matrices at this resolution in the WMAP papers, and no published claim that they overstate the noise; two analyses that fit templates to the same nine-year maps with the same matrices print reduced χ^2 values of 0.96 to 0.98 for V band, and near or above one for Ka and Q, without comment (Lattanzi et al., 2017; Planck Collaboration V, 2020, their Table 1 and Table 12). We use the covariances as released and do not rescale.

The reference frequency. Under flat amplitude priors the answer depends on where the amplitudes are defined. For this band set with a ΛCDM prior on the CMB the pivot lies just above 23 GHz (23.3 GHz, to first order), and a flat prior stated at 23 GHz is within one standard deviation of the profile optimum in every region (−0.9, −0.3, −0.6 and −0.3 sandwich errors). The same prior stated at 30 GHz gives a High Latitude index of −2.696 where the profile likelihood

Table 5: The polarized synchrotron index of the WMAP nine-year K, Ka, Q and V maps (with Planck PR4 353 GHz for dust) in the four regions of Fig. 3, at $N_{\mathrm{side}} = 16$ with the dense noise covariances, all bands brought to a common window, the mask applied, and the ΛCDM covariance as the prior on the CMB (E044). Top: the profile optimum with two errors, the debiased sandwich error, which assumes one index per region, and a block error from the scatter of each region's own score over patches of sky, which does not (none is given for the Spur, where fewer than four patches carry the information); then the answer when the same likelihood is instead marginalized over flat amplitude priors with the synchrotron amplitude defined at 23 or at 30 GHz, or under a zero-mean prior with a free scale. Middle: the difference between the two flat-prior answers, with two first-order predictions of it. Bottom: goodness of fit as χ^2 per element and band, for which two free amplitudes and a CMB amplitude held by its prior leave an expectation of 0.6 on five bands.

	High Latitude	Spur	Intermediate	Plane
elements	2974	266	1698	870
profile likelihood	−3.147	−3.139	−3.091	−3.039
sandwich error	0.029	0.027	0.019	0.015
block error, own scores	0.048		0.028	0.019
effective number of patches	12.9	3.5	13.2	10.3
flat prior at 23 GHz	−3.174	−3.147	−3.102	−3.043
flat prior at 30 GHz	−2.696	−2.993	−2.898	−2.971
zero-mean prior, free scale	−3.251	−3.182	−3.121	−3.043
flat 30 GHz minus flat 23 GHz	+0.478	+0.154	+0.204	+0.072
first order, curvature of the marginal	+0.638	+0.183	+0.238	+0.078
first order, J_c (no flat-prior fit)	+0.692	+0.196	+0.251	+0.080
in sandwich errors	16.4	5.6	10.5	4.8
χ^2 per element and band	0.577	0.591	0.572	0.614
T/J_c	0.047	0.004	0.013	0.004

gives −3.147. Between the two conventions the index moves by +0.478, +0.154, +0.204 and +0.072 in the four regions: 16.4, 5.6, 10.5 and 4.8 standard deviations of the sandwich error, with no change in the data or the model. The move is largest where the amplitudes are many and faint, as Eq. (17) says: the High Latitude region has 2974 of them with a mean squared signal-to-noise ratio J_c/T of 21, the Plane 870 with 231.

The identity of Sect. 4 is exact. What the maps test is its first-order consequence, Eq. (18), in two versions (Table 5). With the curvature of the fitted marginal the measured move is 0.75, 0.84, 0.86 and 0.93 of the prediction. With J_c in its place, a prediction that needs no flat-prior fit at all, it is 0.69, 0.78, 0.81 and 0.90. Both predictions are high in every region, by more where the move is larger: the curvature of the marginal grows away from its optimum, and a first-order statement cannot know that. For moves below 0.1 it is good to 10 percent; at High Latitude, where the move is half a unit of index, it is 33 percent high. We had written these expectations down before the fit; the one for the Spur was too small by a factor of three, because it left out the coupling to the High Latitude region through the shared noise (E044).

The zero-mean prior. With a free scale per region it does not agree with the profile optimum on these regions: it is steeper by −0.104, −0.043, −0.030 and −0.004, that is −3.6, −1.6, −1.5 and −0.2 sandwich errors. For one component and a common noise pattern the two are the same estimator (Sect. 5); here there are three components, the noise pattern differs from band to band, and at High Latitude the amplitudes are faint. In 30 simulations with these regions, this mask and this prior at $N_{\mathrm{side}} = 8$, in which the profile optimum is unbiased, the zero-mean optimum of each realization is steeper than its profile optimum by −1.05, −0.54, −0.44 and −0.22 sandwich errors, the same from one noise realization to the next (scatter 0.01 to 0.02). So it is the zero-mean prior that is biased here, by an amount set by the sky and the noise pattern; the maps at $N_{\mathrm{side}} = 16$ show the same sign and order across the regions, about three times larger in index

Table 6: The fit of Table 5 under three priors on the CMB amplitudes: the ΛCDM covariance, a white prior of 0.2 μK per element, and a flat prior (E044). The index of each region with the profile likelihood, and with flat priors on the synchrotron and dust amplitudes for the two reference frequencies. †: the optimum is at the upper edge of the allowed range, −4.6 to −1.6; the other three flat-prior optima of that row are conditional on it.

	High Latitude	Spur	Intermediate	Plane
profile likelihood, with the sandwich error				
ΛCDM covariance	-3.147 ± 0.029	-3.139 ± 0.027	-3.091 ± 0.019	-3.039 ± 0.015
white, 0.2 μK	-3.173 ± 0.026	-3.137 ± 0.025	-3.104 ± 0.019	-3.040 ± 0.014
flat	-3.041 ± 0.062	-3.122 ± 0.060	-3.017 ± 0.044	-3.038 ± 0.035
flat amplitude priors, synchrotron amplitude defined at 23 GHz				
ΛCDM covariance	−3.174	−3.147	−3.102	−3.043
white, 0.2 μK	−3.190	−3.140	−3.111	−3.043
flat	$-1.600^{\dagger}$	−2.778	−2.414	−2.809
flat amplitude priors, synchrotron amplitude defined at 30 GHz				
ΛCDM covariance	−2.696	−2.993	−2.898	−2.971
white, 0.2 μK	−2.775	−3.046	−2.934	−2.975
flat	$-1.600^{\dagger}$	−2.577	−1.737	−2.519
χ^2 *per element and band (expectation 0.60, 0.60 and 0.40)*				
ΛCDM covariance	0.577	0.591	0.572	0.614
white, 0.2 μK	0.573	0.588	0.570	0.616
flat	0.385	0.408	0.372	0.385

(E044).

The prior on the CMB. The CMB has one amplitude per element like the other components, and the index responds to what that amplitude is allowed to be (Table 6). With a white prior of 0.2 μK per element in place of the ΛCDM covariance the four indices move by −0.026, +0.003, −0.012 and −0.001. What a wrong covariance for the CMB does to the estimate can be computed without a simulation. If the CMB amplitudes have covariance S_t and the fit assumes S_w, then to first order the score keeps its form and acquires a mean and an extra variance that are linear in $\mathsf{S}_\mathrm{t} - \mathsf{S}_\mathrm{w}$ (App. A), so that the optimum has a bias and a covariance

$$\delta\beta = -\mathsf{J}^{-1}\,\mathrm{E}[g], \qquad \mathrm{Cov}(\hat{\beta}) = \mathsf{J}^{-1}\big[\mathsf{J} + \mathsf{T} + \mathsf{q}^{\mathsf{T}}(\mathsf{S}_\mathrm{t} - \mathsf{S}_\mathrm{w})\,\mathsf{q}\big]\mathsf{J}^{-1}, \tag{24}$$

with q the response of the scores to a CMB amplitude through the residual-forming operator of the fit. A unit test with 1500 simulated fits reproduces both (scatter 0.113 and 0.089 predicted 0.113 and 0.091 where the matched formula gives 0.090 and 0.083; bias $+0.016 \pm 0.003$ and $+0.021 \pm 0.002$ predicted +0.021 and +0.020). Evaluated for the white-prior fit of these maps with S_t the ΛCDM covariance: the bias is −0.0023, −0.0004, −0.0009 and −0.0003 (−0.09, −0.02, −0.05 and −0.02 of the quoted error), and the true error is 1.22, 1.21, 1.09 and 1.10 times the quoted one. The white-prior fit quotes an error 3 to 10 percent smaller than the ΛCDM fit does, and on a ΛCDM sky its true error is 5 to 10 percent larger: it looks better and is worse. A white prior of nearly the right total variance is therefore not a harmless stand-in for a ΛCDM sky, in the direction of an error bar that is too small by 9 to 22 percent here; in other configurations the sign can be the other (an independent review found errors too large in one), which is the point of computing it. The same calculation gives the expected difference between the two fits of one sky, a standard deviation of 0.014, 0.012, 0.006 and 0.005: the measured moves, less the bias, are −1.74, +0.26, −1.87 and −0.05 of it. Two of the four are near two and have the same sign; those two indices are correlated at 0.2, and we read the four as consistent with the calculation and no more. An earlier simulation at $N_\mathrm{side} = 8$ on other regions agrees in kind: there the matched prior was calibrated and the white-prior error too small by about a fifth in three of four regions (E040).

With a flat prior on the CMB the profile score is unbiased whatever the CMB is, at the price

of errors 2.12 to 2.36 times larger; the indices are then −3.041, −3.122, −3.017 and −3.038, which is +1.95, +0.33, +1.86 and +0.02 from the ΛCDM version in units of $(\sigma^2_{\text{flat}} - \sigma^2_\Lambda)^{1/2}$, the standard deviation of the difference between a fit and a more efficient fit of the same data. Two regions are again near two, the same two as for the white prior and with the other sign: at High and at Intermediate latitude the index is steeper the more tightly the CMB amplitude is held (−3.173, −3.147 and −3.041 at High Latitude for the white, the ΛCDM and the flat prior; −3.104, −3.091 and −3.017 at Intermediate latitude). Each step is at the edge of what its calculation allows, the three fits share their data, and we have no cause to offer; we report it. The χ^2 of the flat-prior fit, which has exactly two degrees of freedom per element and no prior penalty, is 0.956 of its expectation, 3.4 standard deviations low: the deficit of the central fit is not made by the CMB prior, and as a fluctuation it is now one in a thousand.

A free CMB amplitude does more to the flat-prior marginal than move its pivot (Sect. 4). With flat priors on all three amplitudes the High Latitude index has no optimum inside the allowed range: for a reference of 23 GHz and of 30 GHz alike the fit ends at its upper edge, −1.6, where the profile likelihood of the same model gives −3.041 ± 0.062 (Table 6). At the profile optimum the curvature of the flat-prior marginal is negative in that direction (−366 against an information J of 316): the volume term outweighs the data, which is the case that Sect. 4 describes for further free amplitudes. The index variance carried by one amplitude, $n\sigma^2_\beta$, is 11.4 in this region against 2.5 with the ΛCDM prior, so that a reference a tenth in $\ln\nu$ from the pivot is already a shift of order one. [Scan along the High Latitude index (H11).] The other three flat-prior optima are interior and are 0.23 to 1.44 flatter than the profile optimum for 23 GHz, but they are conditional on a neighbour at the edge. No first-order prediction applies to this row.

Errors that do not assume the model. The sandwich error is the error of the estimator if each region has one index and the noise model is right. An error that does not lean on that comes from the data: at the optimum the score of a region is a sum over elements, $\sum_p g_p = 0$, and under the model the covariance of that sum is $\mathsf{J} + \mathsf{T}$; if the model fails inside the region, the partial sums over patches of sky scatter by more. With the 48 pixels of an $N_{\text{side}} = 2$ grid as patches, $G_B = \sum_{p\in B} g_p$ for one region, the block variance of its index is

$$\widehat{\text{Var}}_{\text{block}}(\hat\beta) = \left[(\mathsf{J}_c^{-1})_{rr}\right]^2 \sum_B \frac{G_B^2}{(1-h_B)^2}, \qquad h_B = \frac{D_B}{\sum_{B'} D_{B'}}, \tag{25}$$

with $D_B = \partial G_B/\partial\beta$ the information carried by patch B (negative values, which occur for patches with almost none, are set to zero). Without the factor $(1-h_B)^{-2}$ this is the cluster sandwich in common use (Liang & Zeger, 1986), which is biased low when a few clusters carry the information; with it, it is the form that in linear regression equals the delete-one-cluster jackknife variance up to a factor $G/(G-1)$ for G clusters and is biased high by $\sum_B h_B/(1-h_B)$ (Bell & McCaffrey, 2002; Cameron & Miller, 2015). Our estimator is not linear, so the correspondence is to first order. We use each region's own scores. The joint form, with the full matrix J_c^{-1} on both sides of the outer products of all regions' scores, lets a region inherit the scatter of a neighbour to which its index is coupled; an independent referee showed that this, and not the region itself, made the largest block error of an earlier version of this analysis.

What a block error means needs care (independent review and referee, E040, E044). It is the error of the estimate as a measurement of the mean index of a population of exchangeable patches; for the information-weighted mean index of the region as realized on this sky, the sandwich error is the right one. It is too small if the index varies coherently over several patches. It is too large, even when the model is true, if the noise covariance or the prior couples patches, because the patch scores are then anticorrelated; so its ratio to the sandwich error under the model has to be taken from simulations with the regions, mask, covariances and prior of the fit. A large ratio says that the model fails somewhere in the region and not how: an index that varies, a compact source with its own spectrum, large-scale noise that the covariance understates, dust

Table 7: Changes of the profile index of Table 5 under analysis choices that have nothing to do with the estimator (E044). "Index from $\ell \leq L$": K band and 353 GHz keep all their modes, which with the prior on the CMB fixes the amplitudes, and Ka, Q and V inform the index through their harmonic coefficients at multipoles 2 to L only. The row "K and Ka only, both fits from $\ell \leq 20$" is the K, Ka fit minus the four-band fit when both are read from multipoles up to 20. The flat-prior row repeats Table 5. In brackets, for the arms that use less of the same data under the same model: the change in units of the standard deviation that the model predicts for it, $(\sigma^2_{\text{arm}} - \sigma^2_{\text{base}})^{1/2}$, and for the white prior in units of the first-order value of Eq. (24). The other arms change the model, and their change has no sampling error. The statistical error of the baseline is given for scale.

	High Latitude	Spur	Intermediate	Plane
baseline: all scales, ΛCDM CMB prior	−3.147	−3.139	−3.091	−3.039
statistical error (sandwich)	0.029	0.027	0.019	0.015
flat amplitude prior at 30 GHz	+0.451	+0.146	+0.193	+0.068
CMB prior: white, 0.2 μK	−0.026 (−1.7)	+0.003 (+0.3)	−0.012 (−1.9)	−0.001 (0.0)
CMB prior: flat	+0.106 (+1.9)	+0.017 (+0.3)	+0.074 (+1.9)	+0.001 (0.0)
index from $\ell \leq 20$	+0.004 (+0.2)	−0.028 (−1.0)	−0.047 (−1.4)	+0.011 (+0.3)
index from $\ell \leq 12$	+0.026 (+0.5)	−0.193 (−0.9)	−0.193 (−1.0)	+0.141 (+1.3)
K and Ka only	+0.012 (+0.6)	−0.005 (−0.2)	+0.037 (+2.4)	+0.025 (+2.0)
dust index 1.48 in place of 1.53	−0.004	−0.005	−0.006	−0.021
dust index 1.58	+0.004	+0.005	+0.006	+0.019
delta bandpasses, nominal frequencies	−0.050	−0.051	−0.048	−0.046

that does not follow one spectrum, residuals of the window matching and leakage of intensity into polarization all scatter the patch scores. And it rests on few patches: the effective number, $(\sum_B D_B)^2 / \sum_B D_B^2$ (an analogue, in shares of the information, of the effective number of clusters of Carter et al. 2017), is 12.9, 3.5, 13.2 and 10.3 here.

On the maps the block error is 1.6, 1.0, 1.4 and 1.3 times the sandwich error in the four regions (Fig. 4). Under the model, in 267 simulations at $N_{\text{side}} = 8$ with these regions, this mask and the ΛCDM prior (synchrotron from the DR1 model smoothed to 5°, noise from the window-matched covariances, a ΛCDM CMB, one index per region), the ratio has a median of 1.34, 1.26, 1.13 and 1.09 and a 97.5th percentile of 1.94, 3.14, 1.67 and 1.76; 38, 178, 35 and 64 of the simulations lie above the map values. In the same simulations the sandwich error is calibrated (standardized errors of standard deviation 1.03, 1.05, 0.95 and 0.97, each known to 0.04; coverage of one standard deviation 0.65, 0.67, 0.71 and 0.73). [The same at $N_{\text{side}} = 16$, a few realizations.] So with the mask the block scores give no evidence that any of these regions is more than one index at the precision of these maps, and the two errors in Table 5 tell the same story within what a block error from a dozen patches can tell.

That was not so before the mask. On the four regions of the earlier version, unmasked, the joint block errors were 2.3 to 9 times the sandwich errors and the χ^2 per element and band was 1.78 and 0.77 in its Galactic Center and Plane. With nothing but the disc around Tau A masked on those regions, the index of its Plane moves from −2.964 to −3.048, six of its statistical errors, and its χ^2 falls to 0.596; we had predicted both before the fit, the referee's first-order estimate of the move being −0.07 (E044). The misfit and the scatter of those fits sat in the emission that the mask removes, and one source carried most of it in one region.

6.4 What moves the index

Table 7 lists what else moves the answer, with the statistical error for scale. Its rows are of two kinds, and they have to be read differently.

Less of the same data. An arm that drops information and keeps the model gives, if the model is right, an estimate that differs from the full one by noise of a known size, $(\sigma^2_{\text{arm}} - \sigma^2_{\text{base}})^{1/2}$; for

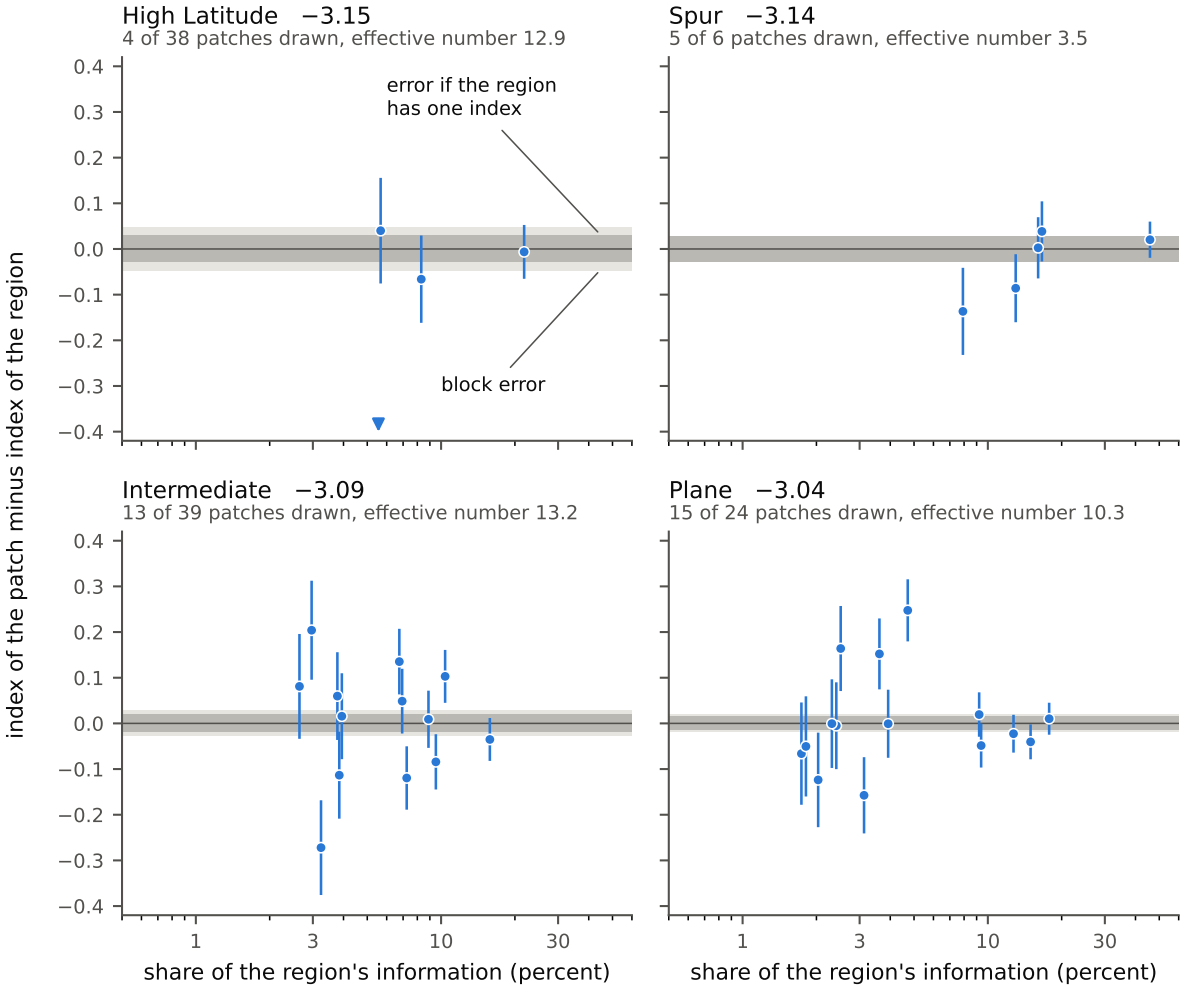


Figure 4: What the block errors of Table 5 are made of. For each patch of sky (a pixel of the $N_{\text{side}} = 2$ grid) that carries information on the index of a region: the index that the patch alone would move to in one Newton step from the fitted index of the region, with the error it would have under the model, against the share of the region's information that the patch carries. Patches with a one-step error above 0.12 are not drawn, and an arrow marks a value outside the frame. Dark band: the sandwich error of the region's index. Light band: its block error (none for the Spur). The fitted index of the region is given beside its name. The one-step values are first-order and for display; the block error is computed from the scores, Eq. (25) (E044).

the estimator that reads the index from large scales this was checked on simulations (E039; [and at $N_{\rm side} = 16$ with these regions]). Such an arm is a test of the model with no free choice in it, and Table 7 gives each change in those units. With the index read from multipoles up to 30, 20 and 12 only, the largest change in any region is [lp30.nest.max pending], 1.4 and 1.3 of them. From K and Ka alone the four indices change by +0.6, −0.2, +2.4 and +2.0, and by [lp20KKa.nest.list pending] when both fits are read from multipoles up to 20. With a flat prior on the CMB they change by +1.95, +0.33, +1.86 and +0.02 (above). The largest of all these is 2.4, in one region of one arm, and the arms share their data. Two patterns are visible and neither is significant: the sequence in the CMB prior described above, and an index from K and Ka alone that is flatter at Intermediate latitude and in the Plane by +0.037 and +0.025, the sign of a spectrum that steepens with frequency.

That is not the picture before the mask. On the earlier regions the index from $\ell \leq 20$ was steeper than the index from all scales by 0.024 to 0.084 in the four regions, 2.8 to 10.4 of the expected scatter, and we could not say why. We had written before the present fits that if compact bright emission was the cause the mask would shrink the largest of them, 0.084 in the old Plane, to between 0.01 and 0.04; in the new Plane the change is +0.011, with the other sign and inside its noise. The dependence on angular scale went away with the emission that the mask removes.

A different model. The other rows change an assumption. Their changes have no sampling error and say what the assumption is worth. A dust index of 1.48 or 1.58 in place of 1.53 moves the synchrotron index by 0.004, 0.005, 0.006 and 0.020, most in the Plane, where it is 1.3 statistical errors. The three fits also say what dust index these maps prefer: $\chi^2_{\rm min}$ is higher by 23.3 at 1.48 and lower by 4.4 at 1.58, and a parabola through the three values has its minimum at 1.57 ± 0.02, against 1.53 ± 0.02 from the Planck bands alone (Planck Collaboration XI, 2020); at that uncertainty the dust index is worth 0.002 to 0.008 in the synchrotron index. This reading of the dust index was not planned and is a by-product. Delta bands at the nominal WMAP frequencies in place of the measured bandpasses steepen every region, by 0.046 to 0.051, which is 1.7 to 3.1 statistical errors: the largest model effect in the table, and the one with no excuse, because the bandpasses are measured. The correction for the drift of the band centres is worth [nodrift.move.abs.max pending] at most, the choice between the polarization-sensitive and all bolometers for the 353 GHz bandpass [bp353full.move.abs.max pending], and the floor of the CMB prior [floor011.move.abs.max pending].

Taken together. For these maps, regions and mask the profile index of the simple model is −3.15, −3.14, −3.09 and −3.04 at High Latitude, in the Spur, at Intermediate latitude and in the Plane, with statistical errors of 0.015 to 0.029. With the mask it does not depend on the angular scale that informs it or on the band subset beyond what the noise allows; it moves by up to 0.106 with the prior on the CMB and by 0.05 if the bandpasses are ignored; and under flat amplitude priors it moves by 0.07 to 0.48 with a convention. We give these numbers as the answer of this model on these regions, with what they are sensitive to, and not as a recommended value of the polarized synchrotron index. An earlier set of regions, without a mask, gave −3.082 to −2.964 with errors of 0.01 for regions that turned out to be mislabelled and misfitted, and it took an independent referee to see it; the choice of regions and mask is itself a model, the spectrum here has no curvature, and the leakage of intensity into polarization is not modelled.

7 Limits: when a proper amplitude prior is needed

The profile likelihood settles the question for a handful of indices that the data determine. It does not settle it for an index field with a hierarchical prior at low signal to noise, where a proper amplitude prior is needed, and there the prior becomes a model component with failure modes

of its own. We list what we established, as limits on what the earlier sections recommend. All of it is on simulations.

A scale shared across index regions on a scale map. If the zero-mean prior takes its scale from a map, $a_p \sim \mathcal{N}(0, (s\,g_p)^2)$ with one s for the sky, a scale map that is wrong differently in different regions biases their indices with a sign that again depends on the reference frequency. On the sky of Table 2, a map that is 30 percent high at $|b| > 30°$ moves the high-latitude indices by -0.026 at a 23 GHz reference, $+0.029$ at 30 GHz and 0.000 at the pivot (`E036`). The effect is coherent over a zone and of the order of its pooled error. One free scale per index region does not have it.

Template-tied priors in intensity. For positive intensity amplitudes we implemented $a_p \sim \mathcal{N}(\mu\,g_p, (s\,g_p)^2)$ with a template g and inferred μ and s per component. This prior passes simulation-based calibration on data drawn from itself (600 replicates, `E016`) and gives calibrated index fields down to index-pixel signal-to-noise ratios below one. It is exact only for its own template-error law. A template error of the same size with a different conditional law, such as a template that is the true amplitude divided by a 30 percent scatter, shifts the faint-sky dust index by 0.11 in the exact target and by 0.13, four posterior standard deviations, in the chain (`E021`): regressing amplitudes on a noisy template is itself an errors-in-variables problem. At high signal to noise with a template built from the data and a sky outside the model class (PySM `d1s1`), the posterior runs away: the index removes the component from all bands but one because that lowers the prior cost by more than it raises χ^2 (`E020`). At the white-noise depth of the WMAP and LFI intensity maps with a 30 percent template error the field mean is within 0.02 of the truth (-3.017 against -3.000, five posterior standard deviations), but the inferred field does not beat a constant index in pixel error on that sky, and the field hyperparameters fail a four-chain split-$\hat{R}$ criterion (1.11 to 1.42) although every chain passes its own convergence check (`E032`).

What follows. An index field with inferred smoothness is a different paper. The conclusion for this one is narrower: where the question is one index per region and the region is measured, use the profile likelihood with the debiased sandwich; do not use a flat amplitude prior unless the reference frequency is the pivot; and do not share one amplitude scale across index regions on a template. None of this makes the index right if the model of the sky or of the instrument is wrong, which is the subject of the last part of Sect. 6.

8 Implementation and validation

Commander4. In our branch of Commander4 the spectral-index Metropolis step of the per-pixel path takes a `likelihood` option. `marginal` evaluates Eq. (4) and `ridge` Eq. (5) in the acceptance, with the amplitudes integrated or maximized analytically per pixel in a compiled kernel; `conditional` is the Gibbs step at fixed amplitudes; `resolve_chisq` keeps the move of Sect. 3 with a warning; `auto` resolves to `ridge` where the kernel applies and to `conditional` otherwise. Each exact option is tested against a grid evaluation of its target in unit tests. End to end, a seven-rank run on simulated maps matched the exact profile posterior within 0.03 posterior standard deviations with width ratios 0.94 and 0.97 (`E006`); in that realization the target itself sat 1.9 and 3.7 standard deviations from the input indices, which a 40-seed repetition showed to be scatter of the harness at its noise level and not an offset (`E037`). On real nside-64 intensity maps the `ridge` chain matched an independent grid evaluation of its target with shifts of -0.06 ± 0.04 and $+0.06 \pm 0.05$ standard deviations and width ratios 0.96 and 0.94 for the two indices (`E014`). On the same maps the conditional Gibbs step has a standard deviation 180 times smaller than the marginal one for the synchrotron index and did not move in 2000 iterations, while the collapsed chains mix in tens of iterations: collapsing the amplitudes is needed for mixing as well as for the

target. Simulation-based calibration of the ridge and marginal modes inside Commander4 itself has not been run.

The low-resolution module. The analysis of Sect. 6 uses a separate, small dense implementation (`nextgen/lowres`): band-averaged spectra from the instrument bandpasses, the collapsed objectives of Sect. 2 for several coupled regions with dense noise covariances, analytic gradients, and J and T of Eq. (6). At $N_{\mathrm{side}} = 16$ with three components the amplitude system has 18 432 unknowns, and a profile fit of four regions takes three to five minutes on a desktop machine. Unit tests check the gradients against finite differences, the diagonal-noise closed forms, J and T against $\sum uV$ and $\sum V$, the sandwich against the scatter of 300 fits, the plug-in bias and its correction, and the identity of Eq. (14) to nine digits. The bandpass readers reproduce the published LFI colour-correction ratios to 3×10^{-4} and the HFI unit conversions to 0.3 percent (E038), and the per-radiometer effective frequencies of the WMAP team's own routine to 0.005 GHz (E039). An independent statistical review of the module, by a reviewer who derived the targets from the stated model before reading our derivations, found no code defect; it corrected the response formula of Sect. 4 (the curvature of the marginal in place of the profile information), the condition under which the profile score is unbiased when a component carries a prior (App. A), and a docstring, and it measured the limits of the plug-in error quoted in Sect. 5 (E039, independent review).

A numerical hazard. During this work three processes crashed with corrupted heaps. The cause was not in our code. On the machine used (Apple M4, macOS 27.0, a pre-release build) the Cholesky routine of the Accelerate framework, called with the upper-triangle option, writes between 0.25 and 0.77 MB past an internal buffer for matrix orders of about 1500 to 6000, and SciPy maps `cho_factor(lower=True)` on a row-ordered array to that option. The factor it returns is correct, so no test of values can see the fault; what lies behind the buffer in memory is overwritten. We found it with the system's guard allocator and measured it with a malloc canary that is now part of the repository, switched the module to the other triangle (same cost, same numbers), and added a test that runs the likelihood under the canary (E043). The simulations at $N_{\mathrm{side}} = 8$ (orders 1536 and 4608) were exposed. Every one of them that this paper uses was run again on the sound path under the canary and compared with the original output as printed: for the 200 realizations of Table 4 all 1480 printed numbers agree to the last digit, as do the 40 realizations with errors from each data set. Fits at $N_{\mathrm{side}} = 16$ (orders 12 288 to 18 432) are outside the range: twenty test matrices at each end of it do not overrun where every one at order 4608 does. All real-data fits reported in this paper were in any case run after the fix, on the sound path under the canary, and none of those runs touched it. We report this because the failure is silent, because dense low-resolution likelihoods of exactly this size are common in this field, and because a result that passes every statistical test can still have been computed next to an overwritten buffer.

9 Discussion and conclusions

What to do. For a spectral index shared by many pixels with linear amplitudes and known noise: (i) evaluate the profile likelihood or a proper marginal directly in the acceptance step, as the marginal and ridge samplers of Andersen et al. (2023) do, never the χ^2 at freshly drawn amplitudes; (ii) do not use a flat amplitude prior unless the amplitudes are defined at the pivot frequency of the band set, which depends on the other components and their priors, and say which reference frequency was used; (iii) quote the profile optimum with the debiased sandwich error, Eq. (22), and check $\mathsf{T}/\mathsf{J}_{\mathrm{c}}$; (iv) if a prior is put on a component that does not depend on the index, state the sensitivity of the index to that prior.

What the real maps show. On the WMAP K to V polarization maps, with regions defined by geometry and the brightest emission masked, changing nothing but the frequency at which a flat amplitude prior is stated, from 23 to 30 GHz, moves the index by 0.07 to 0.48 in the four regions: 4.8 to 16.4 times its statistical error. With a flat prior on the CMB as well the flat-prior marginal has no optimum at high latitude in either convention: the index runs to the edge of the allowed range, -1.6, where the profile likelihood of the same model gives -3.04 ± 0.06. The identity behind this is exact. Its first-order form predicts every move that we could check and is always too high, by 8 to 33 percent with the curvature of the fitted marginal and by 11 to 45 percent with the information of the profile fit alone. The profile likelihood does not depend on the convention. That comparison is differential and does not rest on the sky model being right. Two further results are specific to this configuration. The zero-mean prior with a free scale, which is the same estimator as the profile likelihood for one component and uniform noise, is not here: it lands 3.6, 1.6, 1.5 and 0.2 standard deviations from the profile optimum in the four regions, and in simulations it is the one that is biased. And the prior on the CMB matters beyond the pivot: a white prior of nearly the right total variance in place of the ΛCDM covariance understates the error of the index by 9 to 22 percent on these maps, by a first-order calculation that needs no simulation (Eq. 24).

What they do not show. We have not measured the polarized synchrotron index. The profile index of the simple model is -3.15, -3.14, -3.09 and -3.04 at high latitude, in the Spur, at intermediate latitude and in the plane, with statistical errors of 0.015 to 0.029; with the mask no region is misfitted (the total χ^2 is 2.5 standard deviations below its expectation), and errors from the scatter of the score over patches of sky agree with the model's within what a dozen patches can say. With the mask the index does not depend on the angular scales or the band subset that inform it beyond the noise (the largest change is 2.4 of its expected standard deviations); it moves by up to 0.11 with the prior on the CMB, by 0.020 in the plane with a dust index 0.05 away, and by 0.05 if the bandpasses are ignored. These are the answers of one model on one set of regions, and the route to them is itself a caution. An earlier version of this analysis used four regions reconstructed from a published index map and no mask. Its tables were internally consistent and passed our own checks; its "High Latitude" region was mostly inner Galactic plane, its misfit was the brightest emission, and one source, Tau A, moved the index of one region by 0.08, six of its statistical errors. An independent referee found this from the saved fit products, by asking where on the sky the misfit sat. A χ^2 that is too large is a question about the sky before it is a statement about a model, and a region is a choice that has to be shown. The spread of published values for this sky, -2.9 to -3.3, is of the size of these effects and of the convention effect together, not of the statistical errors.

What we did not do. All results are for maps, at low resolution, with one index per region and Gaussian noise of known covariance. We did not treat time-ordered data, gain or bandpass sampling, or an index field with inferred smoothness, for which a proper amplitude prior is needed and has the failure modes of Sect. 7. The account of the real-map chain in Sect. 3 uses a factor measured from the log of the same chain, and the forward test has not been run. Simulation-based calibration of the exact modes inside Commander4 itself has not been run; they are tested against grids of their targets. The estimator that reads the index from large scales only is unbiased and calibrated on simulations in which the model is true; that says nothing about what it does with whatever makes real maps depend on scale. The block errors treat patches of sky as exchangeable and independent, rest on a dozen effective patches or fewer, are too small if the index varies coherently over several patches, and have to be read against simulations with the covariances of the fit. The ΛCDM prior describes multipoles up to 47. The mask follows a published description and is not a published file; the Spur region rests on a published index map. We did not use the Planck LFI bands: at this precision the official 30 GHz polarization maps do not fit one power law with WMAP K band in our hands, which is the subject of other work.

Reproducibility. Every number in this paper comes from a script in the repository and an experiment record that states the question, the predictions written before the run, the command and the outcome, including the predictions that failed. The records are cited inline.

A Score, curvature and sandwich for the profile likelihood

Let $\Pi = \mathsf{M}\mathsf{A}^{-1}\mathsf{M}^{\mathsf{T}}\mathsf{N}^{-1}$ be the projector onto the model space, so that $\mathsf{P} = \mathsf{N}^{-1}(1-\Pi)$, $\mathsf{P}\mathsf{M} = 0$ and $\chi^2_{\min} = d^{\mathsf{T}}\mathsf{P}d$. With $\mathsf{M}_r = \partial\mathsf{M}/\partial\beta_r$ and $\mathsf{P}_r = \partial\mathsf{P}/\partial\beta_r$, differentiating $\mathsf{P}\mathsf{M} = 0$ gives $\mathsf{P}_r\mathsf{M} = -\mathsf{P}\mathsf{M}_r$. The score is $s_r = -\frac{1}{2}d^{\mathsf{T}}\mathsf{P}_r d$. For $d = \mathsf{M}a + n$:

$$s_r = -\tfrac{1}{2}a^{\mathsf{T}}\mathsf{M}^{\mathsf{T}}\mathsf{P}_r\mathsf{M}a - a^{\mathsf{T}}\mathsf{M}^{\mathsf{T}}\mathsf{P}_r n - \tfrac{1}{2}n^{\mathsf{T}}\mathsf{P}_r n = (\mathsf{M}_r a)^{\mathsf{T}}\mathsf{P}\, n - \tfrac{1}{2}n^{\mathsf{T}}\mathsf{P}_r n, \tag{26}$$

because $\mathsf{M}^{\mathsf{T}}\mathsf{P}_r\mathsf{M} = -\mathsf{M}^{\mathsf{T}}\mathsf{P}\mathsf{M}_r = 0$. The first term has zero mean. The second has mean $-\frac{1}{2}\mathrm{tr}(\mathsf{P}_r\mathsf{N}) = \frac{1}{2}\,\partial\,\mathrm{tr}(\Pi)/\partial\beta_r = 0$, since $\mathrm{tr}\Pi = k$ for every β. So $\mathrm{E}[s_r] = 0$ for any true amplitudes: the profile score is unbiased.

The variance of the first term is $(\mathsf{M}_r a)^{\mathsf{T}}\mathsf{P}\mathsf{N}\mathsf{P}(\mathsf{M}_{r'}a) = (\mathsf{M}_r a)^{\mathsf{T}}\mathsf{P}(\mathsf{M}_{r'}a) = \mathsf{J}_{rr'}$, using $\mathsf{P}\mathsf{N}\mathsf{P} = \mathsf{P}$. The second term is a quadratic form in Gaussian noise, uncorrelated with the first, with covariance $\frac{1}{2}\mathrm{tr}(\mathsf{P}_r\mathsf{N}\mathsf{P}_{r'}\mathsf{N}) = \frac{1}{2}\mathrm{tr}(\Pi_r\Pi_{r'})$. Writing $\mathsf{G} = \mathsf{A}^{-1}\mathsf{M}^{\mathsf{T}}\mathsf{N}^{-1}$,

$$\Pi_r = (1-\Pi)\mathsf{M}_r\mathsf{G} + \mathsf{M}\mathsf{A}^{-1}\mathsf{M}_r^{\mathsf{T}}\mathsf{N}^{-1}(1-\Pi), \tag{27}$$

and since $\mathsf{G}(1-\Pi) = 0$ and $(1-\Pi)\mathsf{M} = 0$ the two squared terms vanish in the trace and the cross terms give

$$\tfrac{1}{2}\mathrm{tr}(\Pi_r\Pi_{r'}) = \mathrm{tr}\big(\mathsf{A}^{-1}\mathsf{M}_r^{\mathsf{T}}\mathsf{P}\,\mathsf{M}_{r'}\big) = \mathsf{T}_{rr'}. \tag{28}$$

The expected curvature follows from differentiating the score once more: $\mathrm{E}[\partial^2(\frac{1}{2}\chi^2_{\min})/\partial\beta_r\partial\beta_{r'}] = \mathsf{J}_{rr'}$. The estimator $\hat\beta$ that solves $s(\hat\beta) = 0$ therefore has covariance $\mathsf{J}^{-1}(\mathsf{J}+\mathsf{T})\mathsf{J}^{-1}$ to leading order, Eq. (19).

A Gaussian prior on a component whose spectrum does not depend on the index (the CMB with variance σ_c^2 per element) fits in by minimizing over that component first: the result is the flat-prior problem for the remaining components with noise covariance $\mathsf{N} + \mathsf{C}\,\Lambda^{-1}\mathsf{C}^{\mathsf{T}}$, with C the columns of the constrained component and Λ its prior precision. Everything above then holds in expectation over noise and over skies of that component drawn from the prior, with its sample variance counted in $\mathsf{J}+\mathsf{T}$. It does not hold for a prior variance that differs from the true one. If the true variance is κ^2 times the prior variance, the expected gradient of $\frac{1}{2}\chi^2_{\min}$ is $(\kappa^2-1)\,t$ (the expected score is minus that) with $t_r = \mathrm{tr}(\Lambda\mathsf{A}^{-1}\mathsf{M}_r^{\mathsf{T}}\mathsf{B}^{\mathsf{T}})$, $\mathsf{B} = \mathsf{A}^{-1}\mathsf{M}^{\mathsf{T}}\mathsf{N}^{-1}$, A containing the prior precision, and the optimum is biased by $-(\kappa^2-1)\,\mathsf{J}^{-1}t$ to first order. A single factor κ^2 describes a prior that is wrong in size; a prior that is wrong in shape, as a white prior is for the CMB, needs the calculation mode by mode. Because t_r is a sum over elements of one sign, the bias in units of the standard deviation grows as the square root of the number of elements.

For a prior that is wrong in shape the same calculation gives Eq. (24). Write $\mathsf{F} = [\mathsf{M}\ \mathsf{C}]$ for all columns, $\mathsf{A} = \mathsf{F}^{\mathsf{T}}\mathsf{N}^{-1}\mathsf{F}$ plus $\Lambda = \mathsf{S}_{\mathrm{w}}^{-1}$ on the block of the constrained component, $\hat a = \mathsf{G}d$ with $\mathsf{G} = \mathsf{A}^{-1}\mathsf{F}^{\mathsf{T}}\mathsf{N}^{-1}$, and $r = \mathsf{R}d$ with $\mathsf{R} = 1 - \mathsf{F}\mathsf{G}$ for the residual. The gradient of $\frac{1}{2}\chi^2_{\min}$ is $g_r = -(\mathsf{F}_r\hat a)^{\mathsf{T}}\mathsf{N}^{-1}r$ with $\mathsf{F}_r = \partial\mathsf{F}/\partial\beta_r$. With data $d = \mathsf{M}a + \epsilon$, $\epsilon = \mathsf{C}c + n$ and $\mathrm{Cov}(c) = \mathsf{S}_{\mathrm{t}}$,

$$g_r = -(\mathsf{F}_r a)^{\mathsf{T}}\mathsf{N}^{-1}\mathsf{R}\,\epsilon - \epsilon^{\mathsf{T}}\mathsf{G}^{\mathsf{T}}\mathsf{F}_r^{\mathsf{T}}\mathsf{N}^{-1}\mathsf{R}\,\epsilon. \tag{29}$$

Both terms have the expectations of the matched case when $\mathsf{S}_{\mathrm{t}} = \mathsf{S}_{\mathrm{w}}$ (zero mean, variance $\mathsf{J}+\mathsf{T}$). The difference is linear in $\Delta = \mathsf{S}_{\mathrm{t}} - \mathsf{S}_{\mathrm{w}}$:

$$\mathrm{E}[g_r] = -\mathrm{tr}\big[\Delta\,(\mathsf{F}_r\mathsf{Z})^{\mathsf{T}}\,\mathsf{N}^{-1}\mathsf{R}\mathsf{C}\big], \qquad \mathrm{Cov}(g_r, g_{r'}) \simeq \mathsf{J}_{rr'} + \mathsf{T}_{rr'} + \mathsf{q}_r^{\mathsf{T}}\Delta\,\mathsf{q}_{r'}, \qquad \mathsf{q}_r = (\mathsf{N}^{-1}\mathsf{R}\mathsf{C})^{\mathsf{T}}\,\mathsf{F}_r a, \tag{30}$$

with $\mathsf{Z} = \mathsf{G}\mathsf{C} = \mathsf{E}_{\mathrm{c}} - \mathsf{A}_{\cdot\mathrm{c}}^{-1}\Lambda$ the response of the fitted amplitudes to a unit amplitude of the constrained component (E_{c} selects its block). The mean is exact to the order of the sandwich; the

covariance neglects the change of the noise-noise term under the mismatch. The optimum is then biased by $-\mathsf{J}^{-1}\mathrm{E}[g]$ and has covariance $\mathsf{J}^{-1}[\mathsf{J}+\mathsf{T}+\mathsf{q}^{\mathsf{T}}\Delta\mathsf{q}]\mathsf{J}^{-1}$. Every matrix in these expressions is formed in the fit; on real data J is replaced by J_{c} and a by $\hat{a}$.

For the fitted amplitudes $\hat{a} = a+e$ with $\mathrm{Cov}(e) = \mathsf{A}^{-1}$, $\mathrm{E}[(\mathsf{M}_r\hat{a})^{\mathsf{T}}\mathsf{P}(\mathsf{M}_{r'}\hat{a})] = \mathsf{J}_{rr'}+\mathrm{tr}(\mathsf{A}^{-1}\mathsf{M}_r^{\mathsf{T}}\mathsf{P}\mathsf{M}_{r'}) = \mathsf{J}_{rr'} + \mathsf{T}_{rr'}$, which gives Eq. (22).

For one power-law component, delta bands and diagonal noise, $\mathsf{M}_r a$ has elements $a_p m_b \ln(\nu_b/\nu_{\mathrm{ref}})$, the projector removes the component of $\ln\nu$ along the amplitude direction, and $\mathsf{J} = \sum_p a_p^2 A_p V_p$, $\mathsf{T} = \sum_p V_p$ with $A_p = \sum_b w_{pb} m_b^2$ and V_p the weighted variance of $\ln\nu_b$ under the weights $w_{pb}m_b^2$, Eq. (7).

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